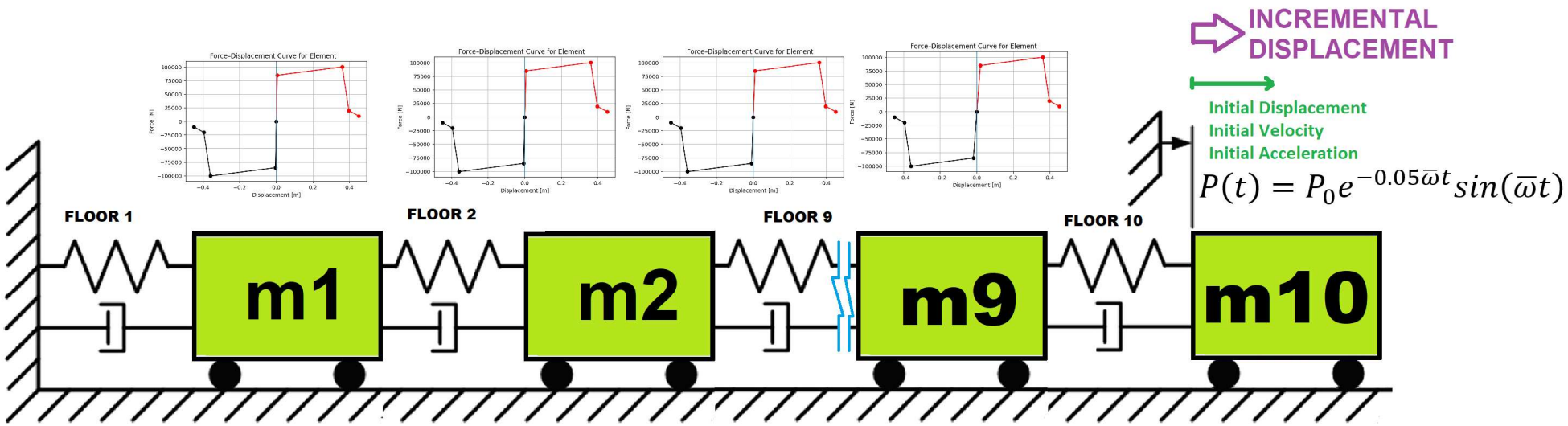
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

PARAMETRIC NONLINEAR SEISMIC DEMAND SPECTRA OF A MULTI-DEGREE-OF-FREEDOM (MDOF) SYSTEMS WITH VARIABLE DUCTILITY CAPACITY

WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

This OpenSees script performs a nonlinear MDOF column ductility response-spectrum analysis that characterizes an oscillator's hysteretic behavior, energy dissipation, and damage under earthquake excitation. It defines mass, stiffness, yield strength, ultimate displacement, and damping; applies a cyclic displacement protocol to establish baseline energy dissipation; then runs nonlinear time-history analyses with Newmark integration over a parametric sweep of periods. For each period, it extracts peak displacement, velocity, acceleration, and base reaction, and computes the Ductility Damage Index, Energy Dissipation Capacity Index, and Equivalent Viscous Damping Ratio. Finally, it statistically processes and plots displacement, velocity, acceleration, base-reaction, DDI, EDCI, and EVDR spectra, generates fragility information, and exports results for performance-based earthquake engineering and parametric studies.

Since the structure was assumed to be inelastic and the response spectra were then plotted for the energy dissipation capacity index and the ductility damage index, this shows the natural period beyond which an increase in the structure's period leads to a higher probability of damage, and as the ductility ratio of the structure increases, the level of damage decreases.





$$Structural\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

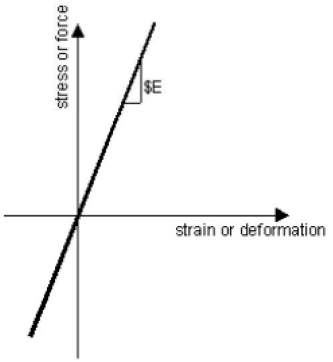
$\Delta_d = Lateral\ Displaement\ from\ Dynamic\ Analysis$

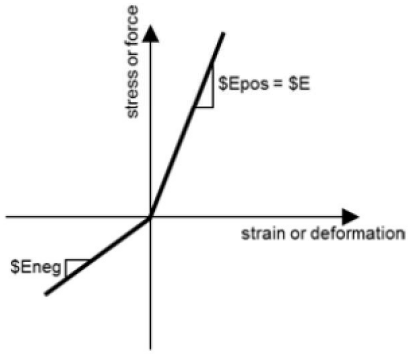
$\Delta_y = Lateral\ Yield\ Displaement\ from\ Pushover\ Analysis$

$\Delta_u = Lateral\ Ultimate\ Displaement\ from\ Pushover\ Analysis$

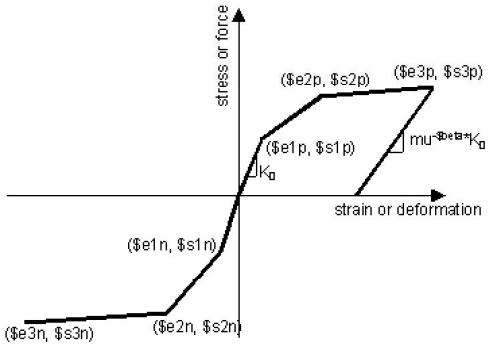
$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$

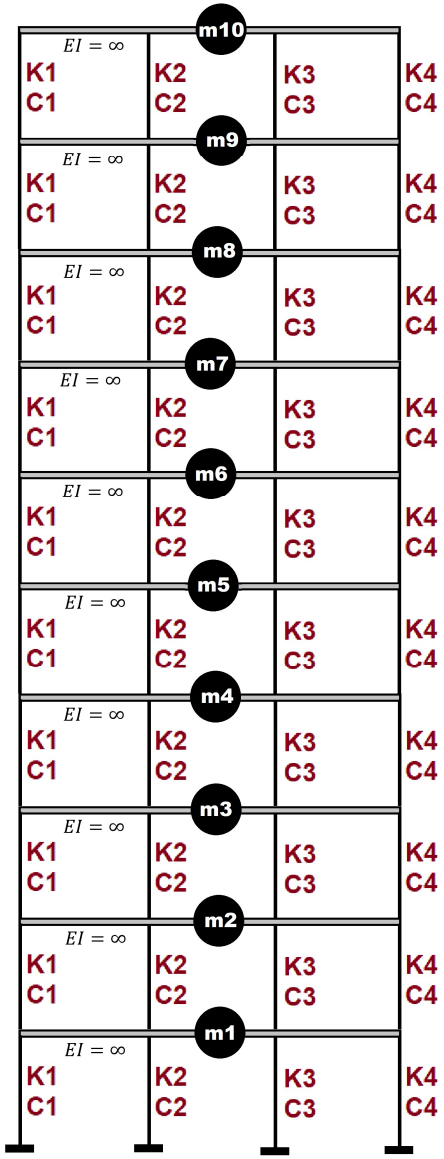




ELASTIC MATERIAL



INELASTIC MATERIAL


m10
m9
m8
m7
m6
m5
m4
m3
m2
m1
EI = ∞
K1
C1
K2
C2
K3
C3
K4
C4

$$Structural\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

$$\Delta_d = Lateral\ Displaement\ from\ Dynamic\ Analysis$$

$$\Delta_y = Lateral\ Yield\ Displaement\ from\ Pushover\ Analysis$$

$$\Delta_u = Lateral\ Ultimate\ Displaement\ from\ Pushover\ Analysis$$

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...DCI_EVDR\MDOF_DUCT_RESPONSE_SPECTRUM_EDCI_EVDR.py

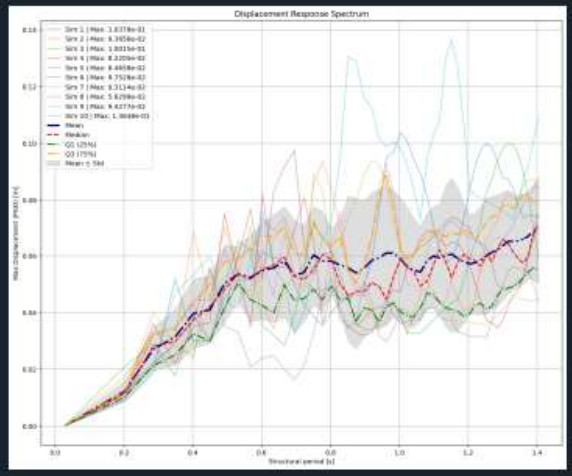
MDOF_DUCT_RESPONSE_SPECTRUM_EDCI_EVDR.py | EQULIVALENT_VISCOUS_DAMPING_RATIO_FUN.py

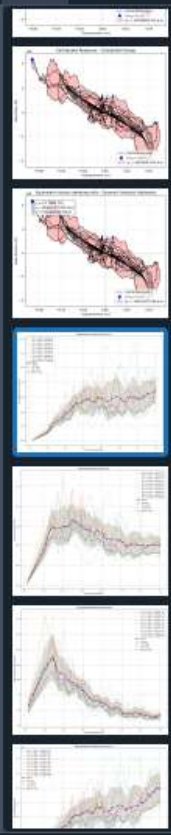
```
##########################################################################################################
#                    >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
#          PARAMETRIC NONLINEAR SEISMIC DEMAND SPECTRA OF A MULTI-DEGREE-OF-FREEDOM (MDOF) SYSTEMS
#                                  WITH VARIABLE DUCTILITY CAPACITY
#---------------------------------------------------------------------------------------------------------
#                     INELASTIC RESPONSE SPECTRA FOR VARYING COLUMN DUCTILITY CAPACITY
#---------------------------------------------------------------------------------------------------------
#                 EQUIVALENT VISCOUS DAMPING RATIO: xi_eq = 100 * E_d / (4 * pi * E_s)
#---------------------------------------------------------------------------------------------------------
#          ENERGY DISSIPATION CAPACITY INDEX = 100 * E_d(earthquake) / E_d(cyclic displacement)
#---------------------------------------------------------------------------------------------------------
# This program performs  Dynamic Analysis on a Multi-Degree-of-Freedom (MDOF) system
# subjected to 1 seismic ground motion. The analysis evaluates the structural response under varying
# levels of seismic intensity.
# The framework is designed to support researchers and engineers in assessing the probabilistic seismi
# performance of structures, with a focus on understanding the impact of uncertainty on structural
# response and design.
#---------------------------------------------------------------------------------------------------------
# Key Features:
# - Simulation of MDOF system using OpenSees.
# - Scaling of ground motions for Nonlinear Dynamic Analysis.
# - Probabilistic fragility assessment based on predefined damage states.
# - Visualization of structural response and fragility curves.
# - Export of results for further analysis.
#---------------------------------------------------------------------------------------------------------
#                      THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
#                                   EMAIL: salar.d.ghashghaei@gmail.com
##########################################################################################################

"""
This OpenSees script performs a nonlinear MDOF ductility response-spectrum analysis
 that characterizes an oscillator's hysteretic behavior, energy dissipation, and
 damage under earthquake excitation. It defines mass, stiffness, yield strength,
 ultimate displacement, and damping; applies a cyclic displacement protocol to
```

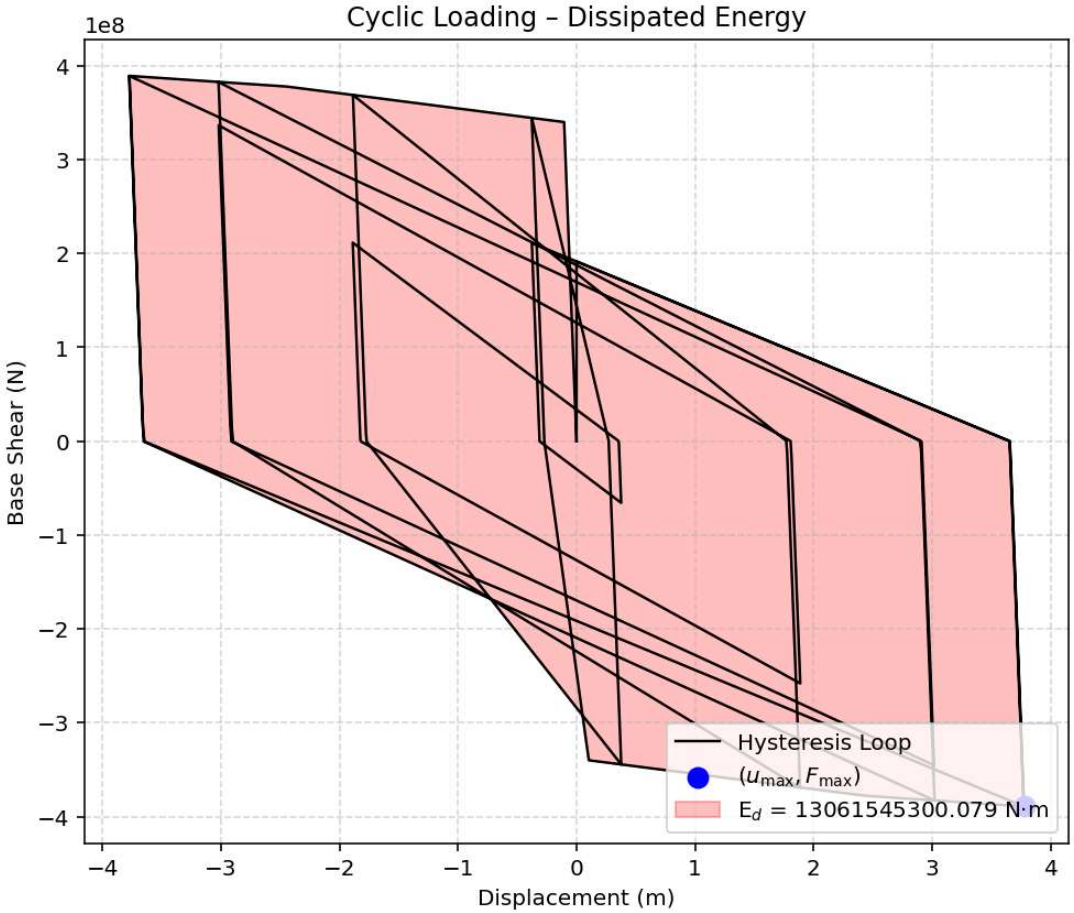
...S_04_COLUMNS_8_ANA\MDOF_DUCT_RESPONSE_SPECTRUM_EDCI_EVDR

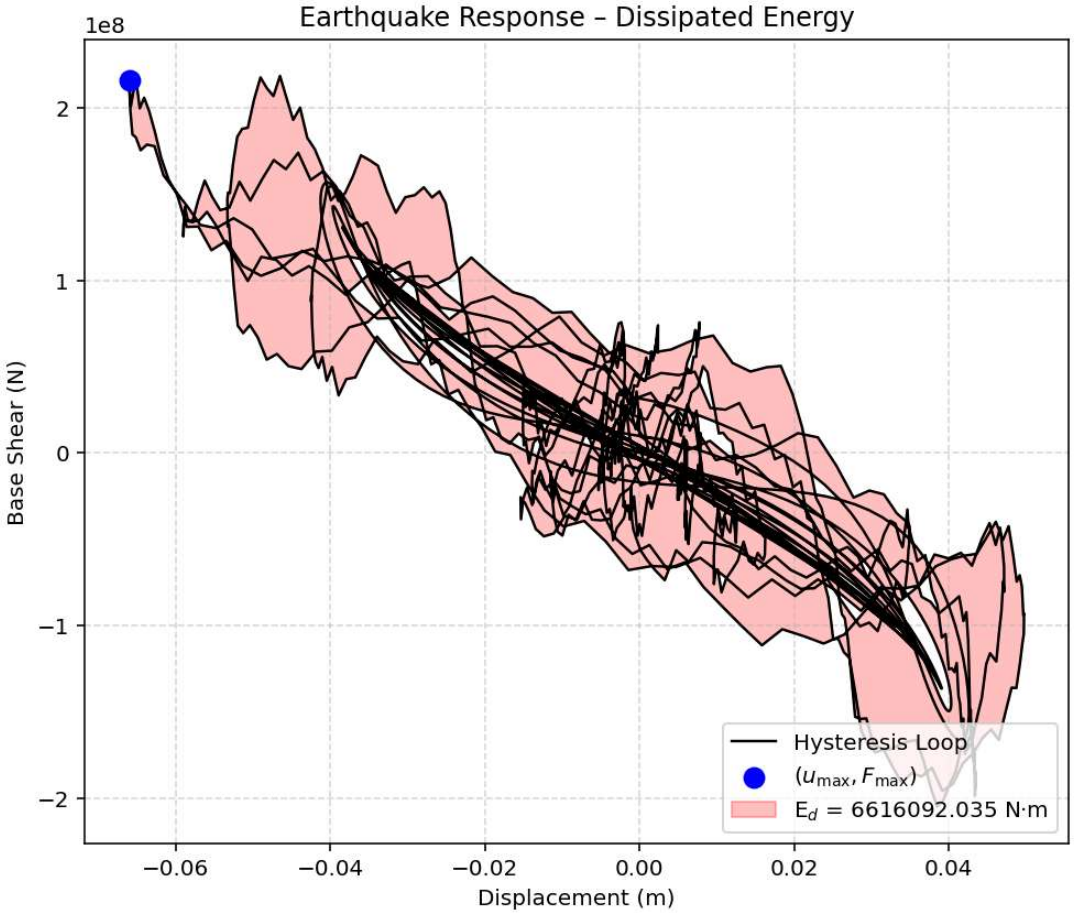
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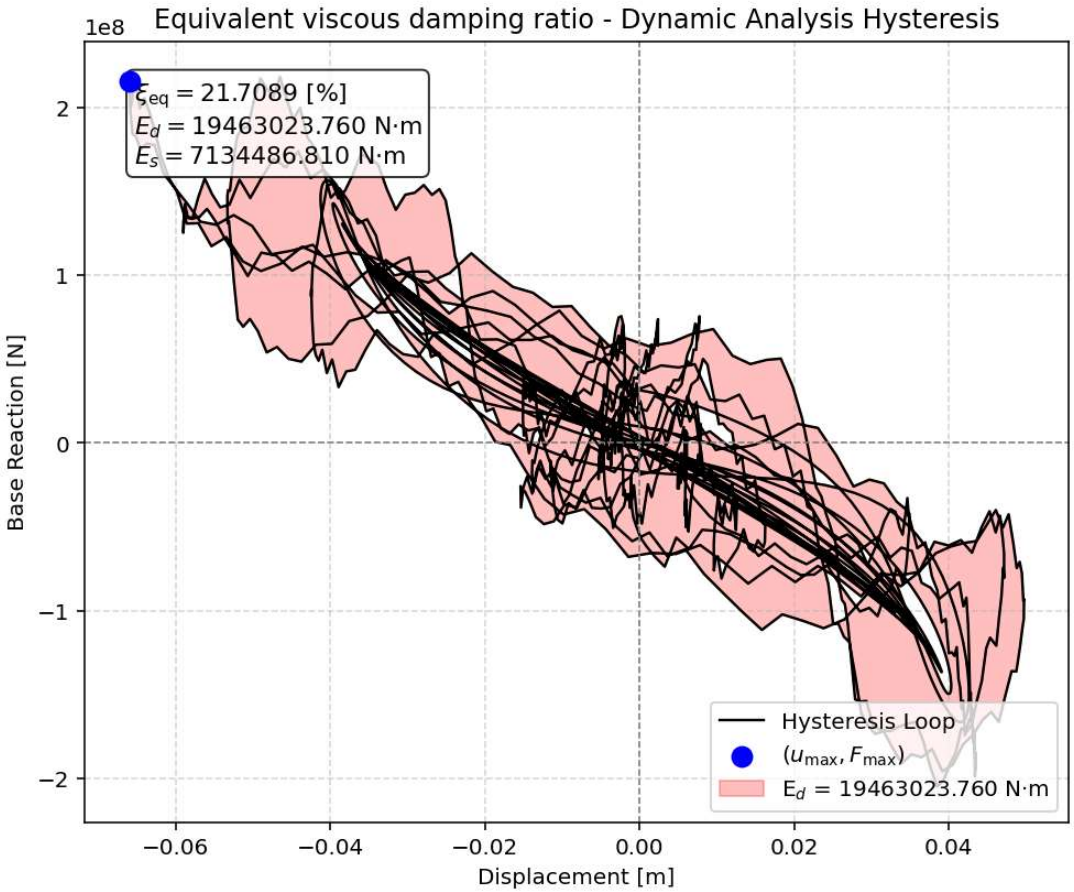


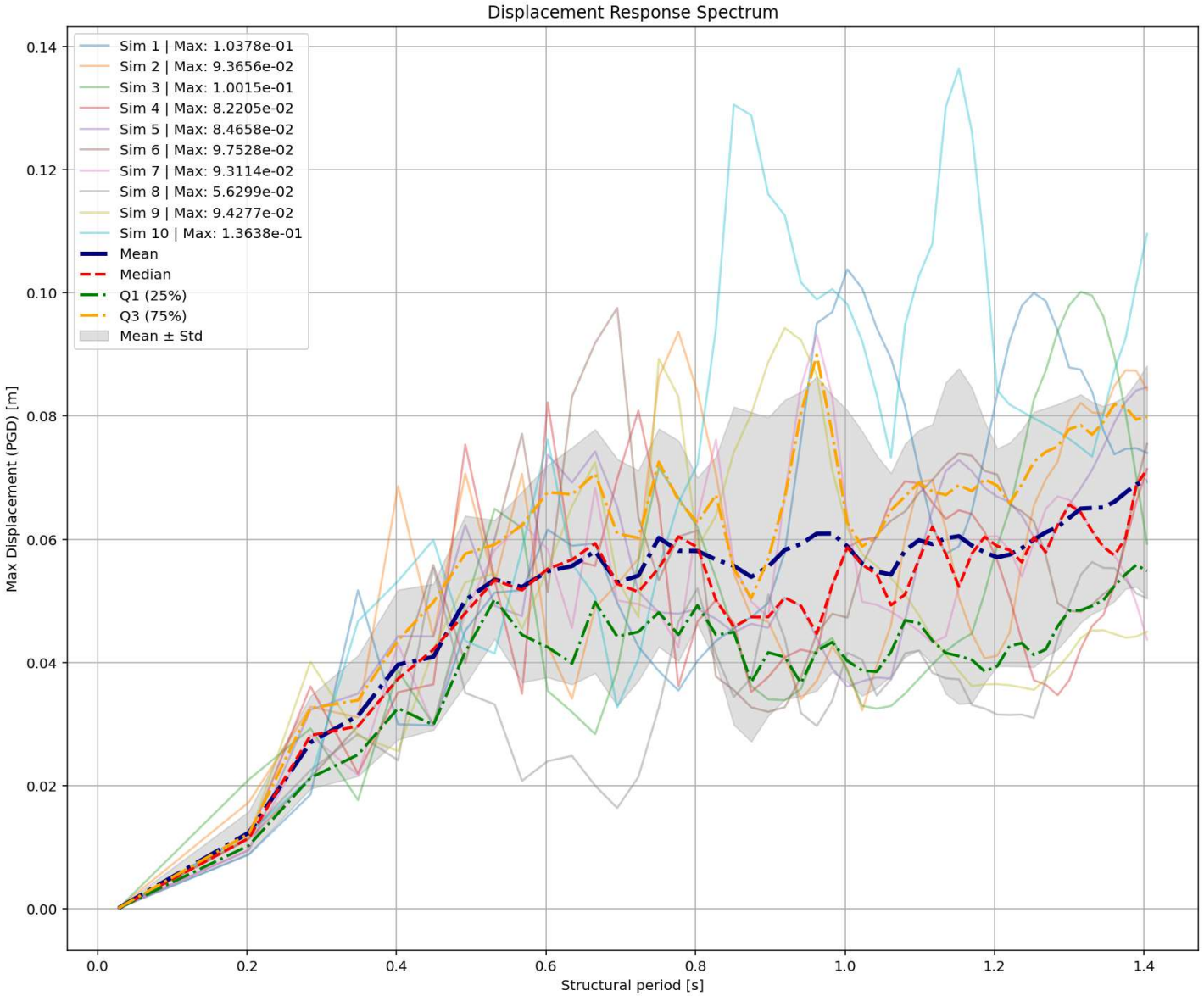


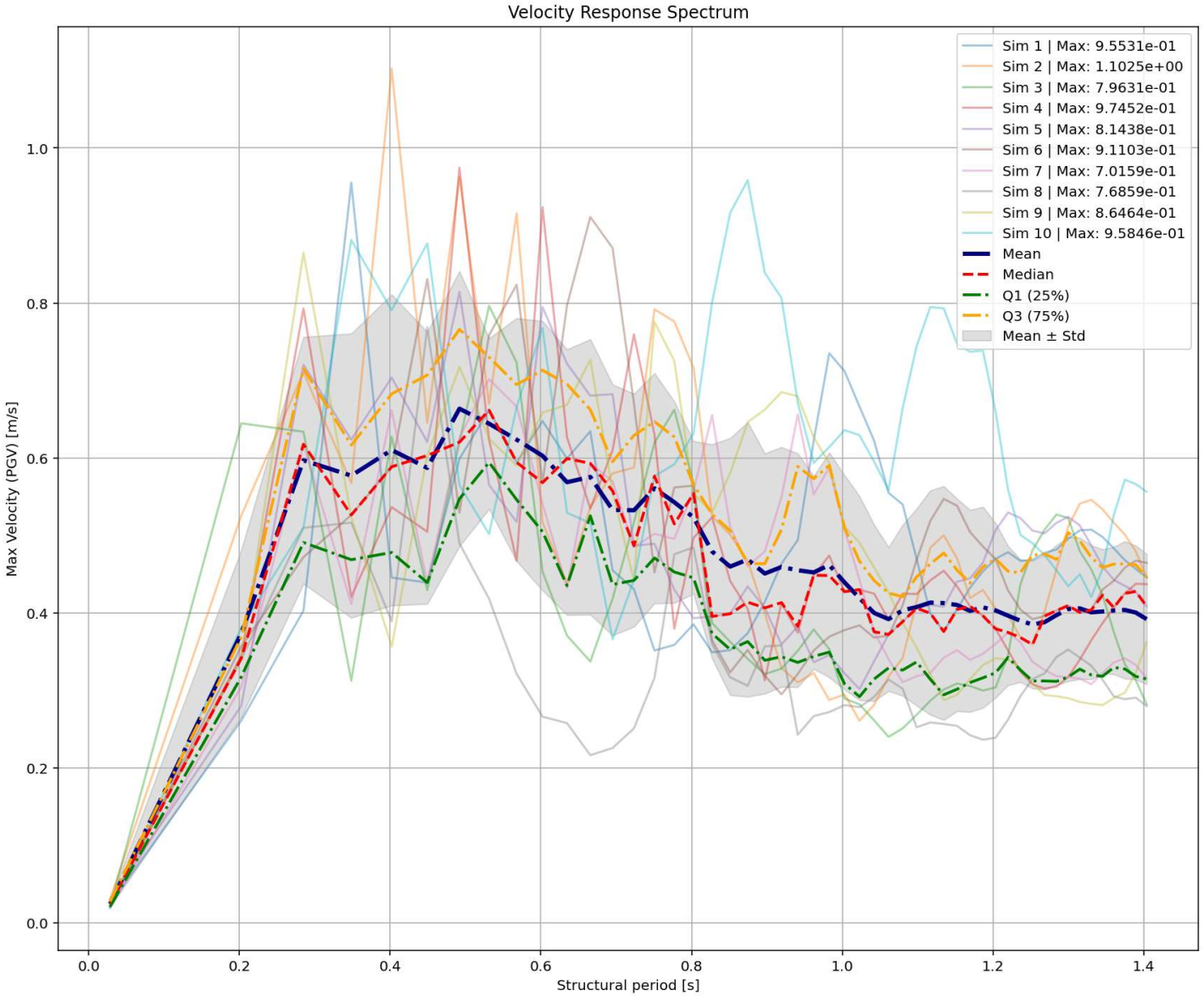
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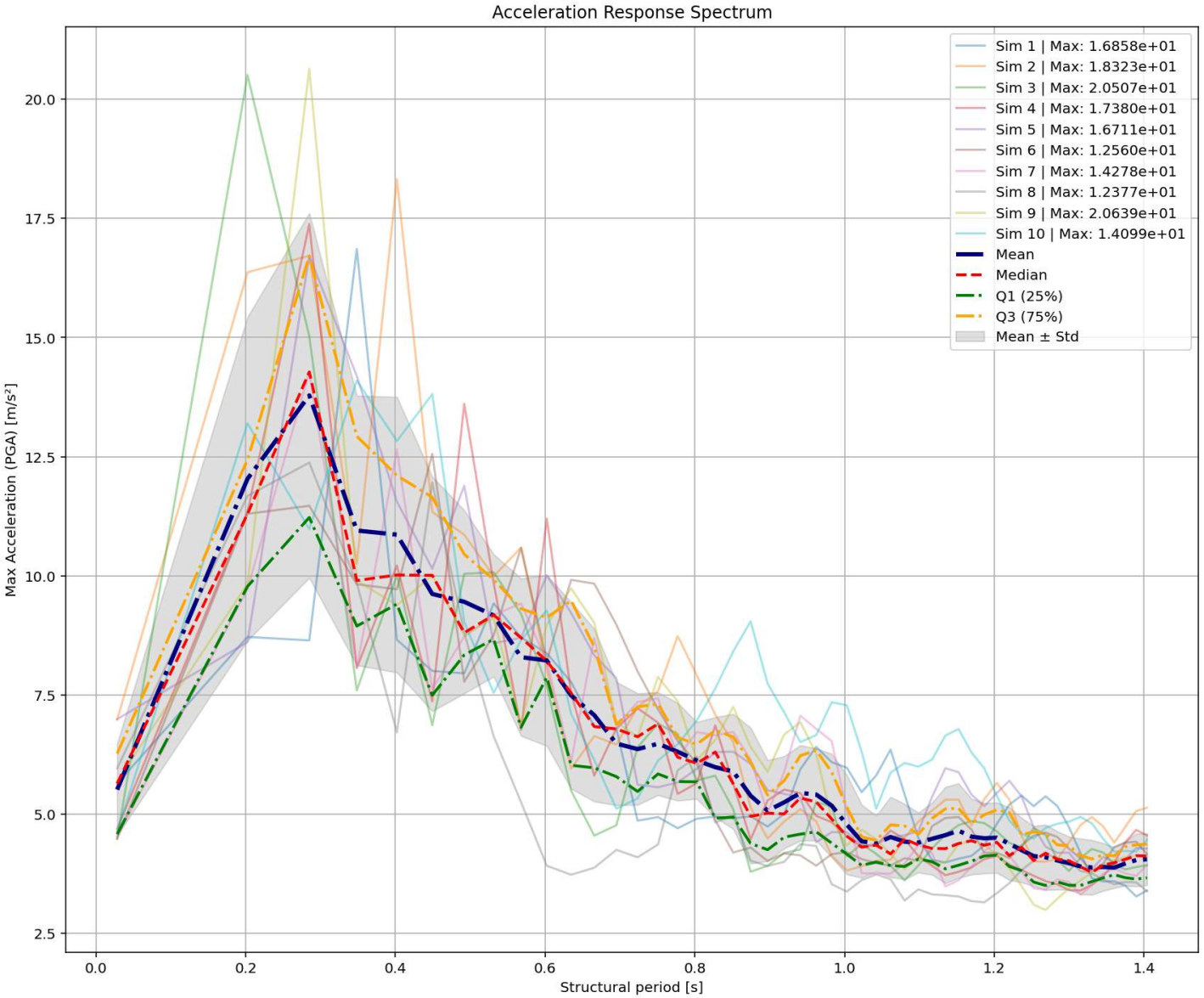

Cyclic Loading – Dissipated Energy
1e8
Base Shear (N)
Displacement (m)
Hysteresis Loop
(u_{max}, F_{max})
E_d = 13061545300.079 N·m

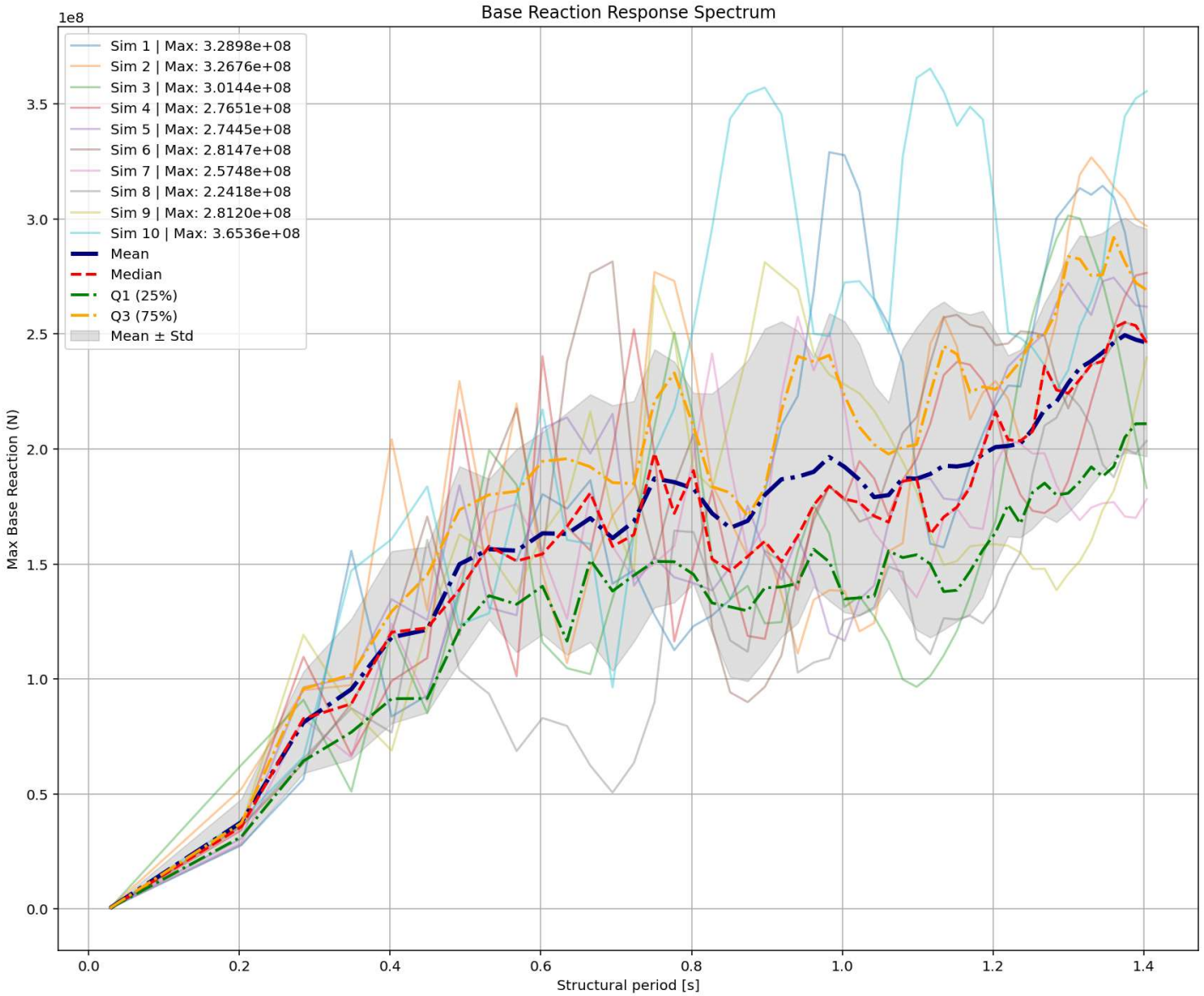

Earthquake Response – Dissipated Energy
1e8
2
1
0
−1
−2
Base Shear (N)
−0.06
−0.04
−0.02
0.00
0.02
0.04
Displacement (m)
Hysteresis Loop
(u_{max}, F_{max})
E_d = 6616092.035 N·m

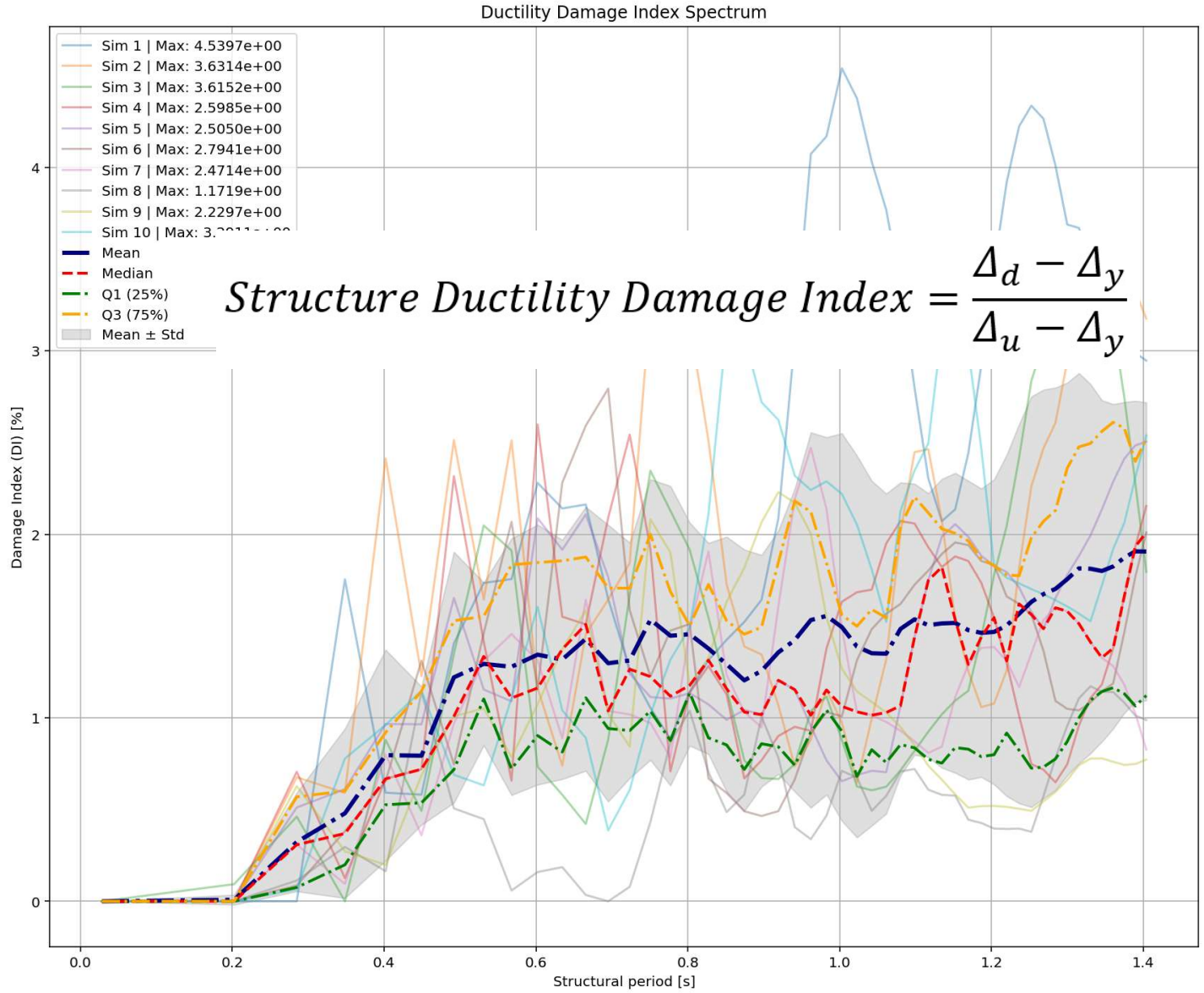

Equivalent viscous damping ratio - Dynamic Analysis Hysteresis
1e8
$\xi_{eq} = 21.7089$ [%]
$E_d = 19463023.760$ N·m
$E_s = 7134486.810$ N·m
Base Reaction [N]
Displacement [m]
2
1
0
−1
−2
−0.06
−0.04
−0.02
0.00
0.02
0.04
Hysteresis Loop
(u_{max}, F_{max})
E_d = 19463023.760 N·m

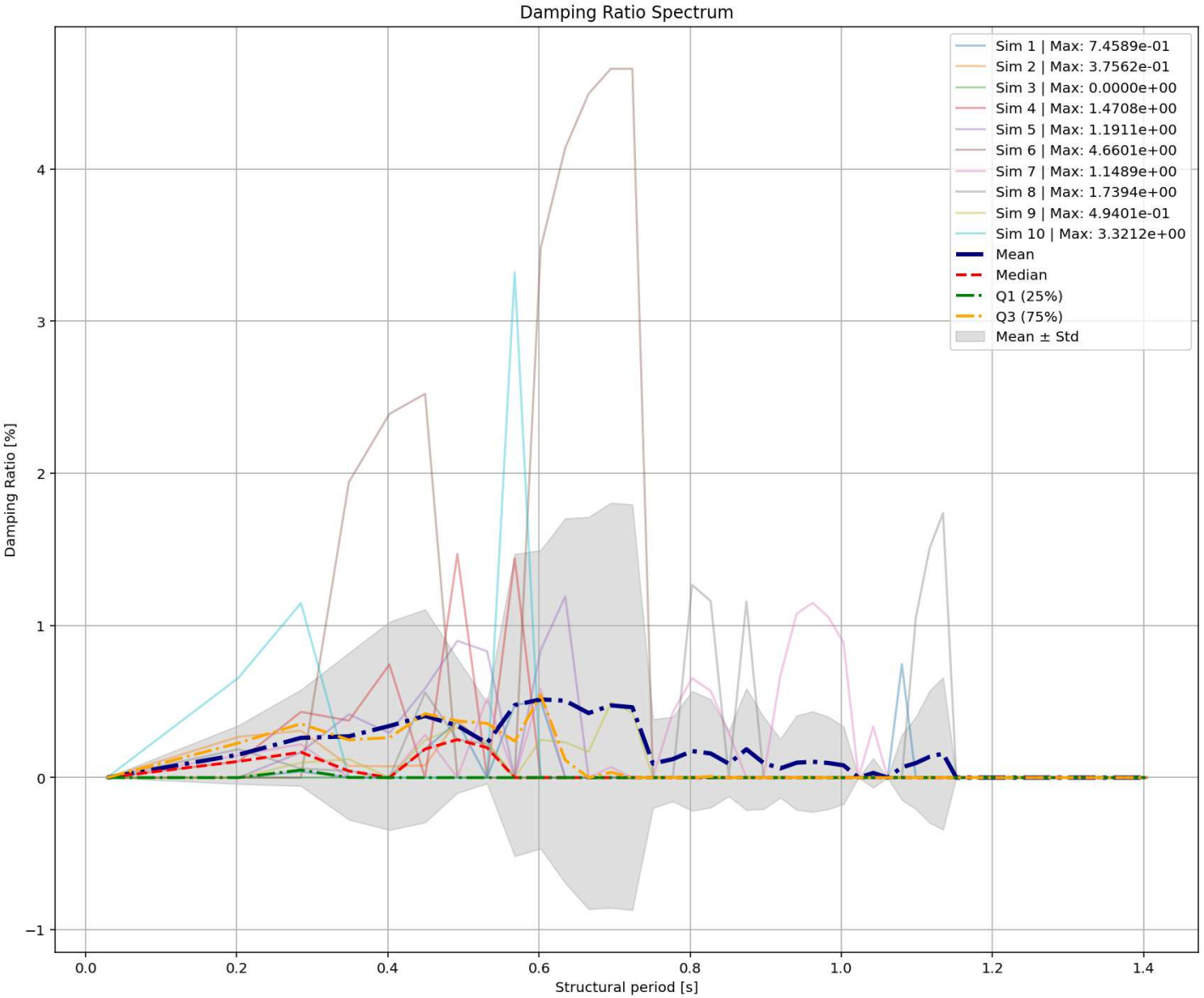

Displacement Response Spectrum
Sim 1 | Max: 1.0378e-01
Sim 2 | Max: 9.3656e-02
Sim 3 | Max: 1.0015e-01
Sim 4 | Max: 8.2205e-02
Sim 5 | Max: 8.4658e-02
Sim 6 | Max: 9.7528e-02
Sim 7 | Max: 9.3114e-02
Sim 8 | Max: 5.6299e-02
Sim 9 | Max: 9.4277e-02
Sim 10 | Max: 1.3638e-01
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Max Displacement (PGD) [m]
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Structural period [s]


Velocity Response Spectrum
Sim 1 | Max: 9.5531e-01
Sim 2 | Max: 1.1025e+00
Sim 3 | Max: 7.9631e-01
Sim 4 | Max: 9.7452e-01
Sim 5 | Max: 8.1438e-01
Sim 6 | Max: 9.1103e-01
Sim 7 | Max: 7.0159e-01
Sim 8 | Max: 7.6859e-01
Sim 9 | Max: 8.6464e-01
Sim 10 | Max: 9.5846e-01
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Max Velocity (PGV) [m/s]
Structural period [s]
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4

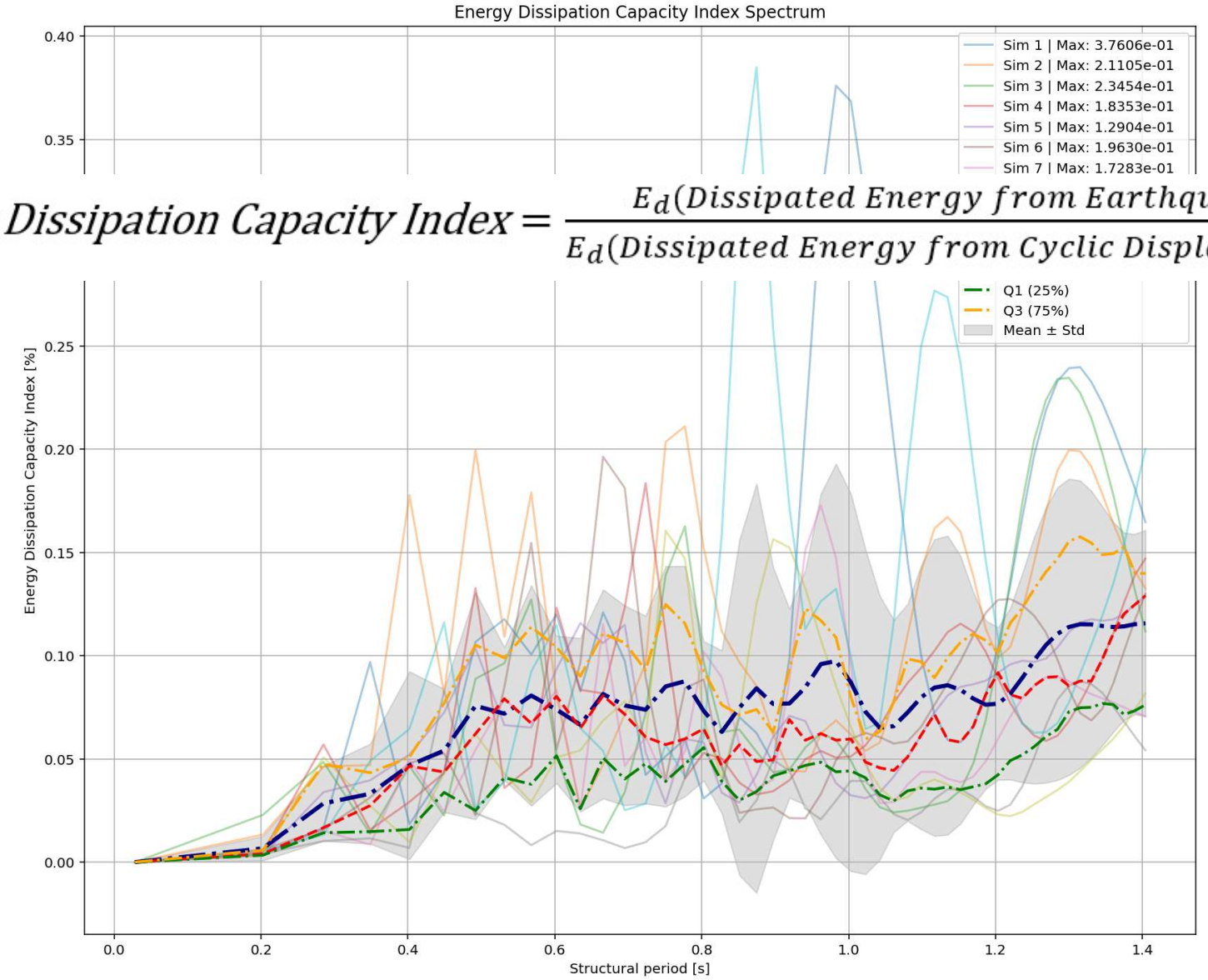

Acceleration Response Spectrum
Sim 1 | Max: 1.6858e+01
Sim 2 | Max: 1.8323e+01
Sim 3 | Max: 2.0507e+01
Sim 4 | Max: 1.7380e+01
Sim 5 | Max: 1.6711e+01
Sim 6 | Max: 1.2560e+01
Sim 7 | Max: 1.4278e+01
Sim 8 | Max: 1.2377e+01
Sim 9 | Max: 2.0639e+01
Sim 10 | Max: 1.4099e+01
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Max Acceleration (PGA) [m/s²]
Structural period [s]
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4

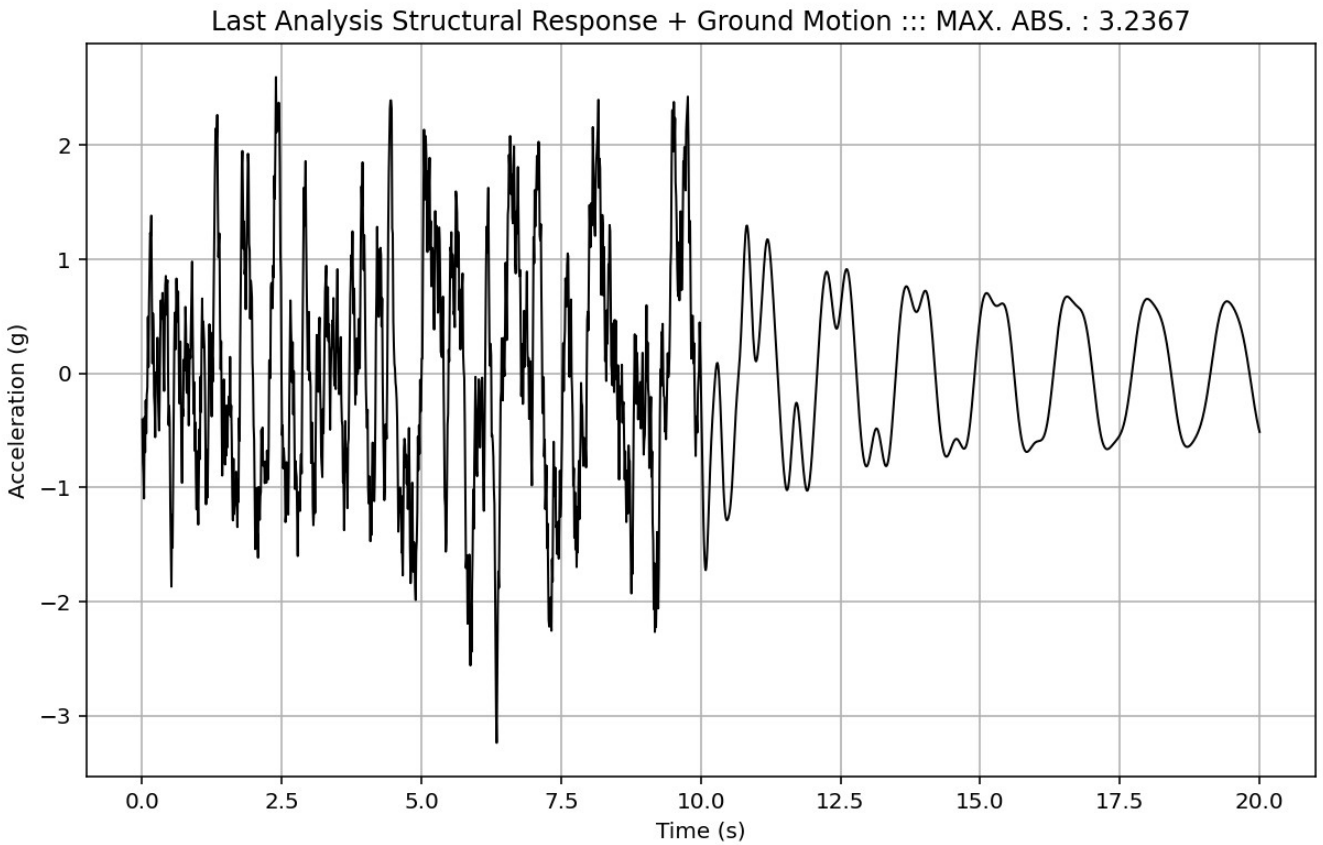

Base Reaction Response Spectrum
1e8
Sim 1 | Max: 3.2898e+08
Sim 2 | Max: 3.2676e+08
Sim 3 | Max: 3.0144e+08
Sim 4 | Max: 2.7651e+08
Sim 5 | Max: 2.7445e+08
Sim 6 | Max: 2.8147e+08
Sim 7 | Max: 2.5748e+08
Sim 8 | Max: 2.2418e+08
Sim 9 | Max: 2.8120e+08
Sim 10 | Max: 3.6536e+08
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Max Base Reaction (N)
Structural period [s]
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4

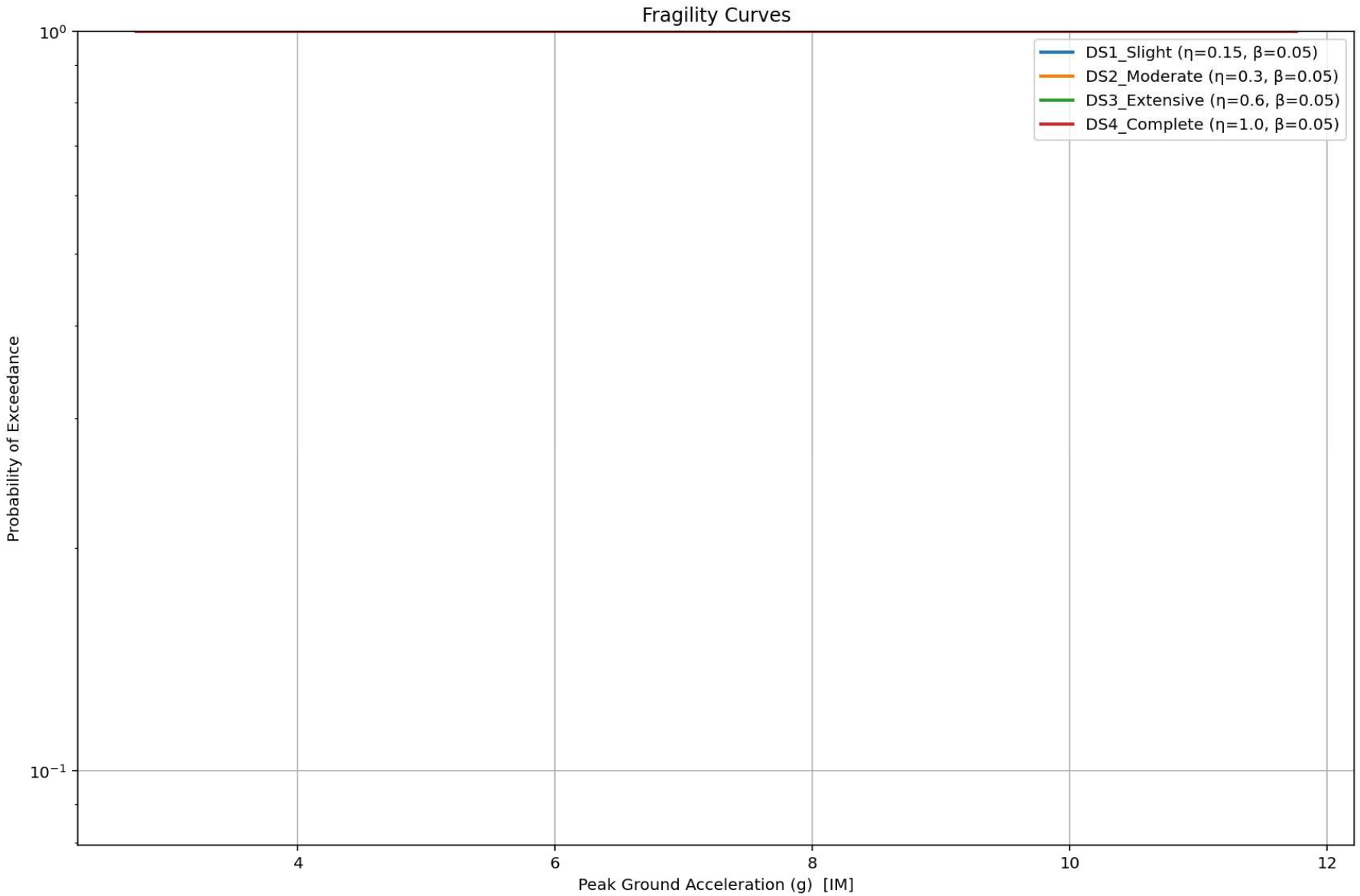

Ductility Damage Index Spectrum
Sim 1 | Max: 4.5397e+00
Sim 2 | Max: 3.6314e+00
Sim 3 | Max: 3.6152e+00
Sim 4 | Max: 2.5985e+00
Sim 5 | Max: 2.5050e+00
Sim 6 | Max: 2.7941e+00
Sim 7 | Max: 2.4714e+00
Sim 8 | Max: 1.1719e+00
Sim 9 | Max: 2.2297e+00
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
$Structure\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$
Damage Index (DI) [%]
Structural period [s]
0
1
2
3
4
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4

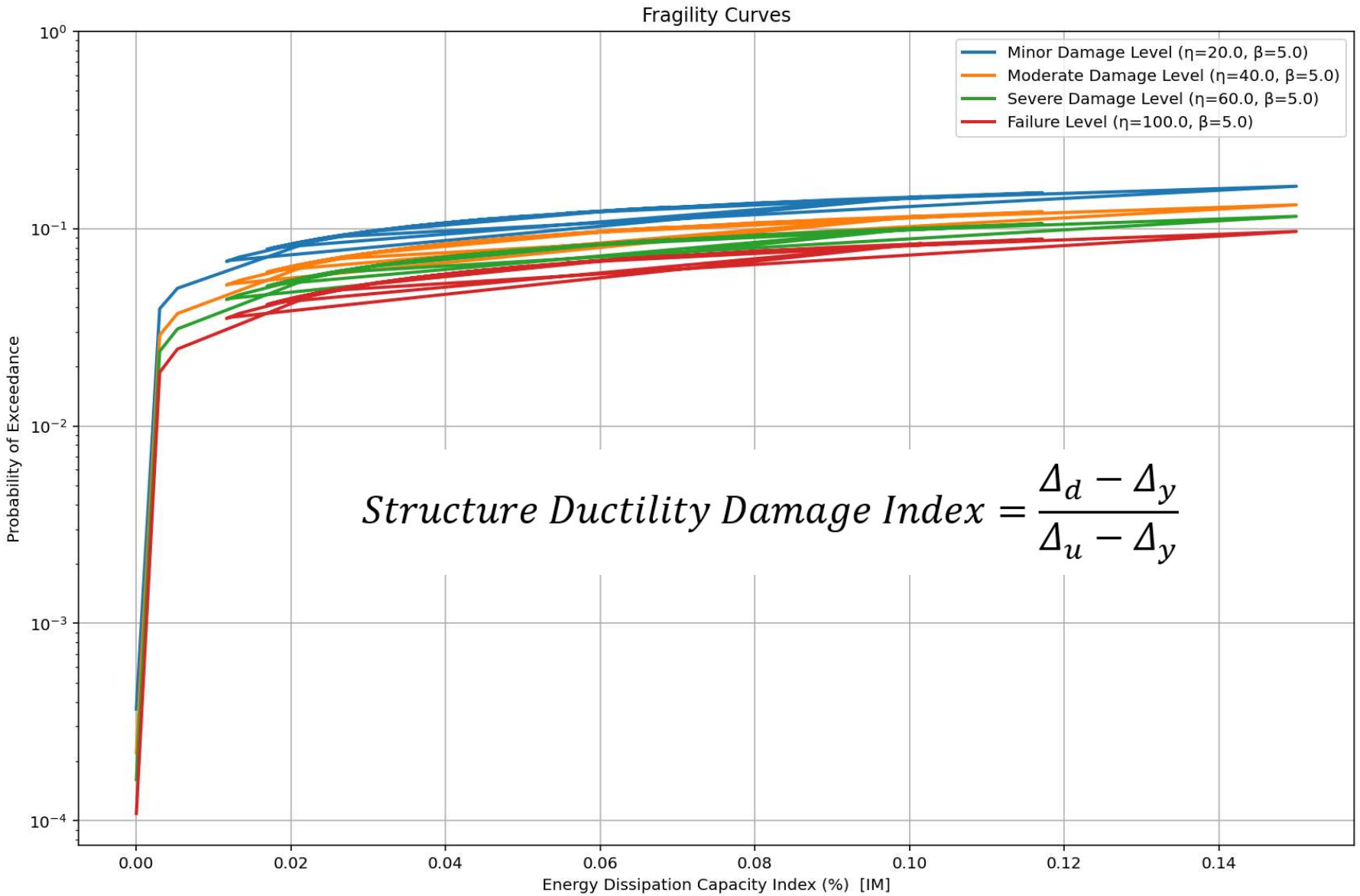

Damping Ratio Spectrum
Sim 1 | Max: 7.4589e-01
Sim 2 | Max: 3.7562e-01
Sim 3 | Max: 0.0000e+00
Sim 4 | Max: 1.4708e+00
Sim 5 | Max: 1.1911e+00
Sim 6 | Max: 4.6601e+00
Sim 7 | Max: 1.1489e+00
Sim 8 | Max: 1.7394e+00
Sim 9 | Max: 4.9401e-01
Sim 10 | Max: 3.3212e+00
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Damping Ratio [%]
Structural period [s]
4
3
2
1
0
−1
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4

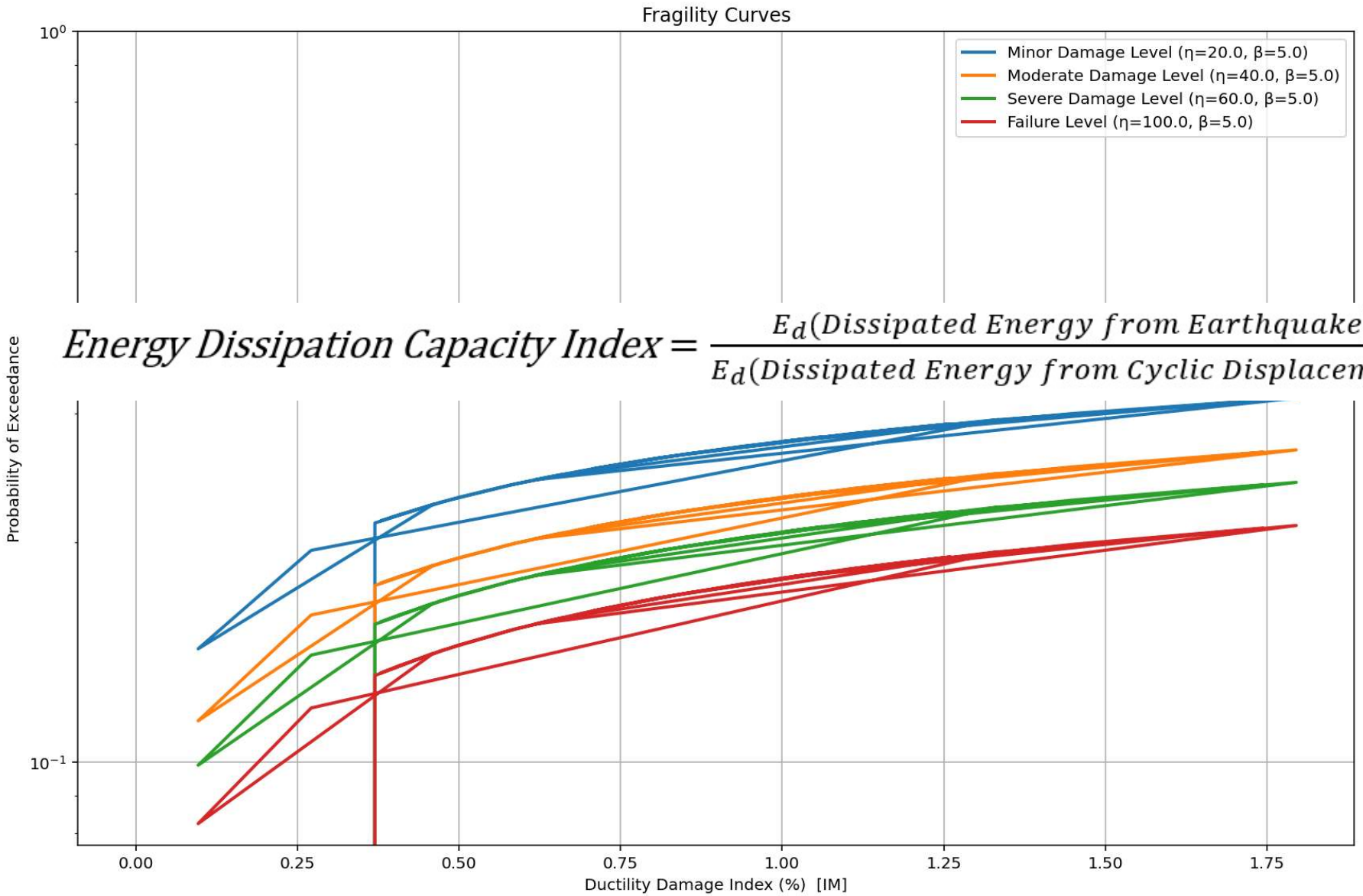
$$Energy\ Dissipation\ Capacity\ Index = \frac{E_d(Dissipated\ Energy\ from\ Earthquake)}{E_d(Dissipated\ Energy\ from\ Cyclic\ Displacement)} * 100$$



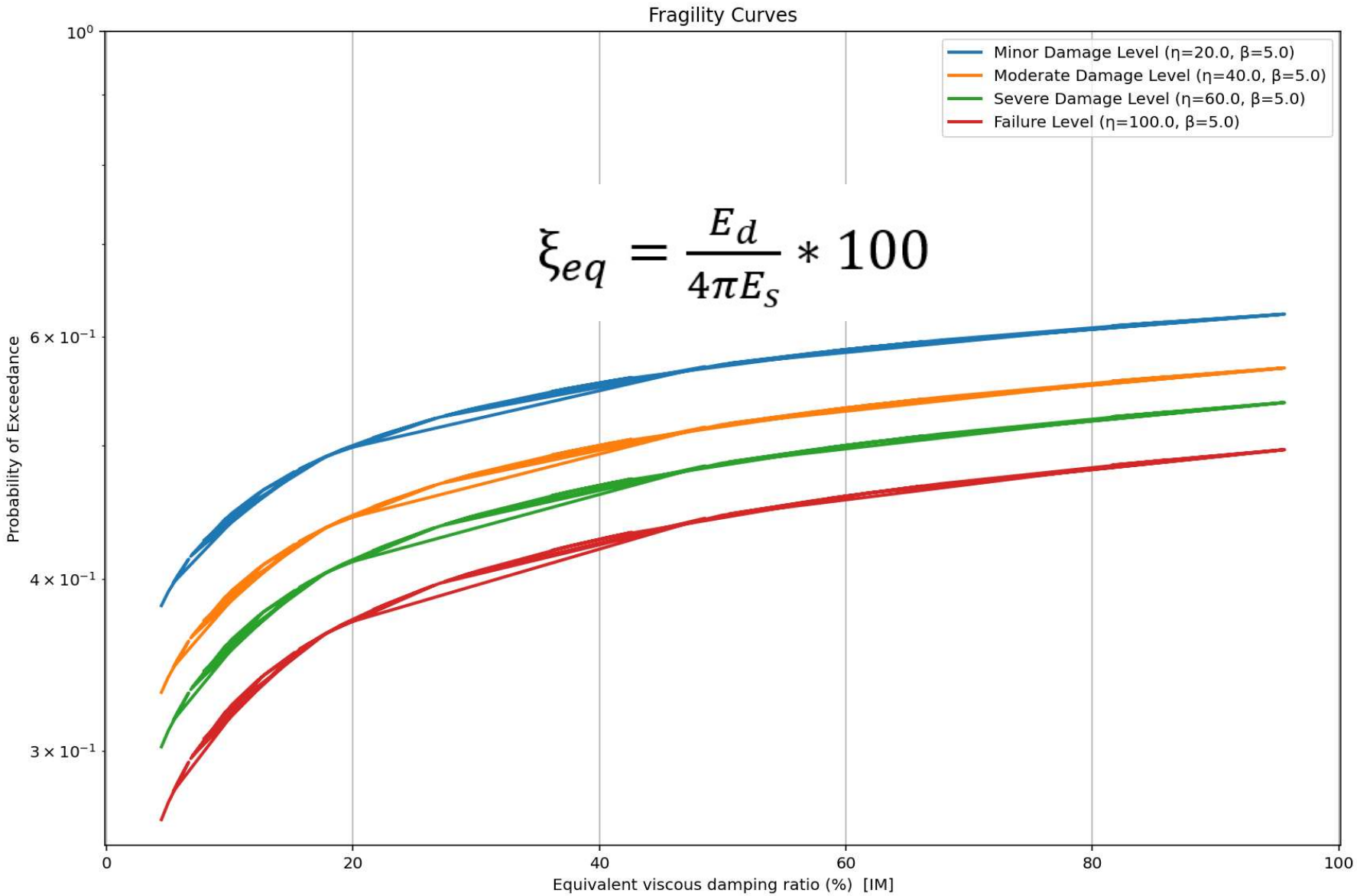

Last Analysis Structural Response + Ground Motion ::: MAX. ABS. : 3.2367
Acceleration (g)
2
1
0
−1
−2
−3
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

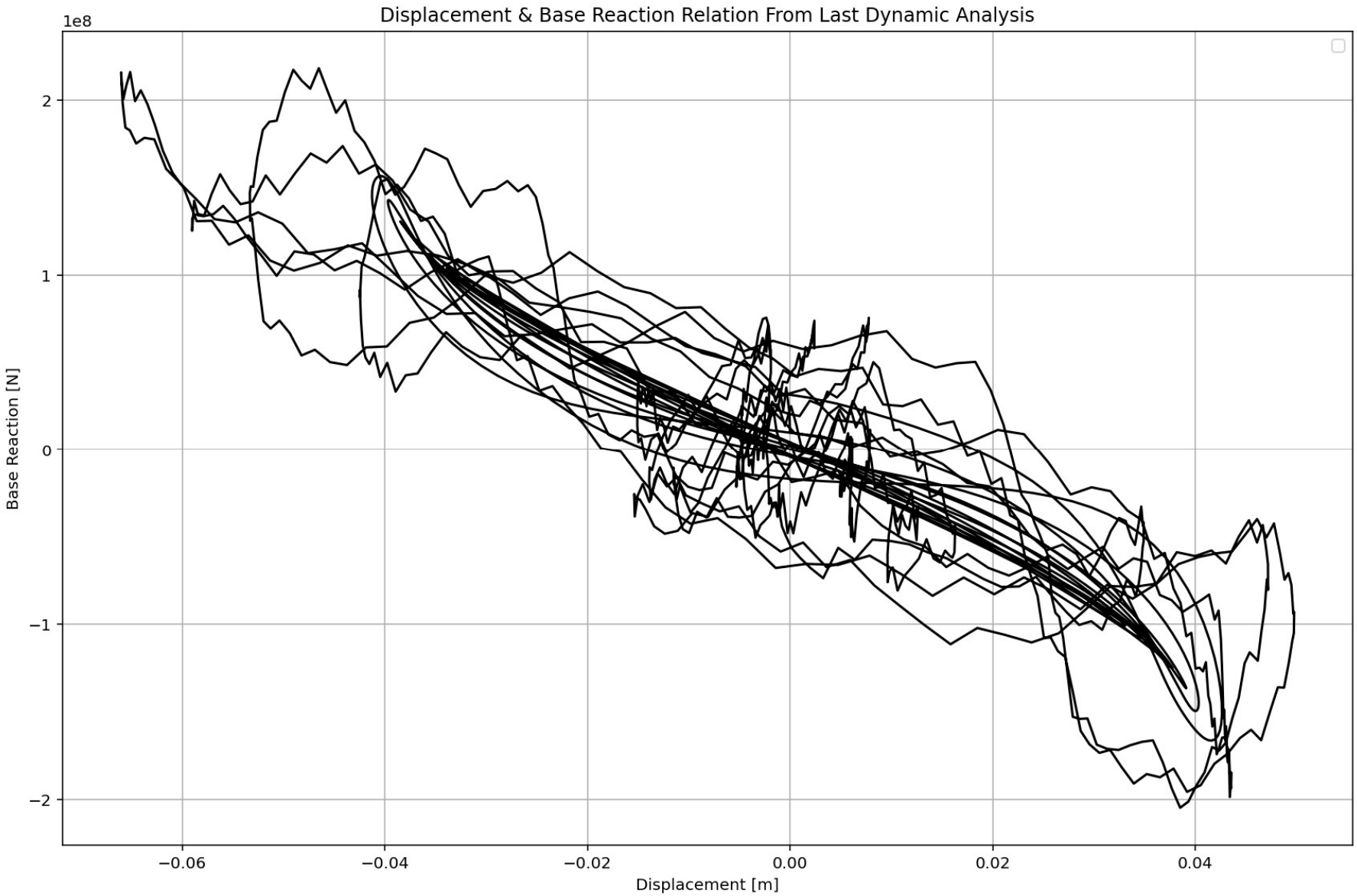

Fragility Curves
DS1_Slight (η=0.15, β=0.05)
DS2_Moderate (η=0.3, β=0.05)
DS3_Extensive (η=0.6, β=0.05)
DS4_Complete (η=1.0, β=0.05)
Probability of Exceedance
10^0
10^{-1}
4
6
8
10
12
Peak Ground Acceleration (g) [IM]

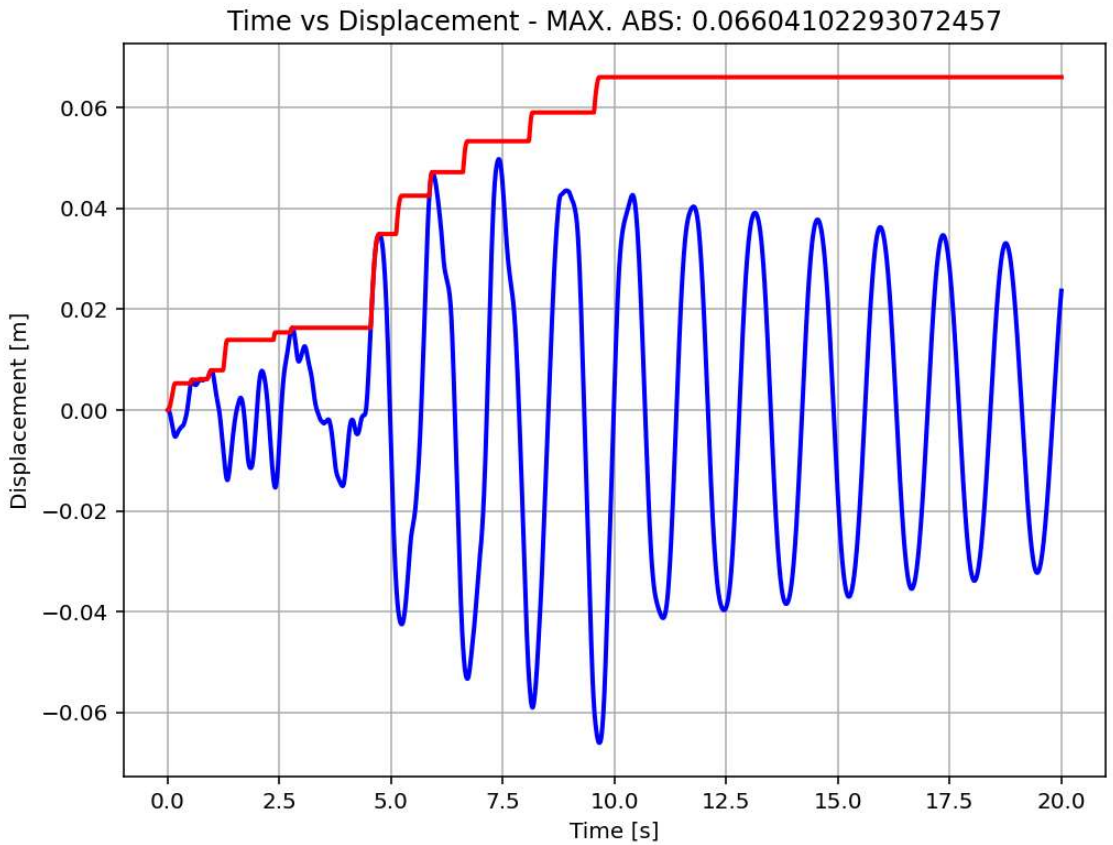

Fragility Curves
Minor Damage Level (η=20.0, β=5.0)
Moderate Damage Level (η=40.0, β=5.0)
Severe Damage Level (η=60.0, β=5.0)
Failure Level (η=100.0, β=5.0)
Probability of Exceedance
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
Energy Dissipation Capacity Index (%) [IM]
$Structure\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$

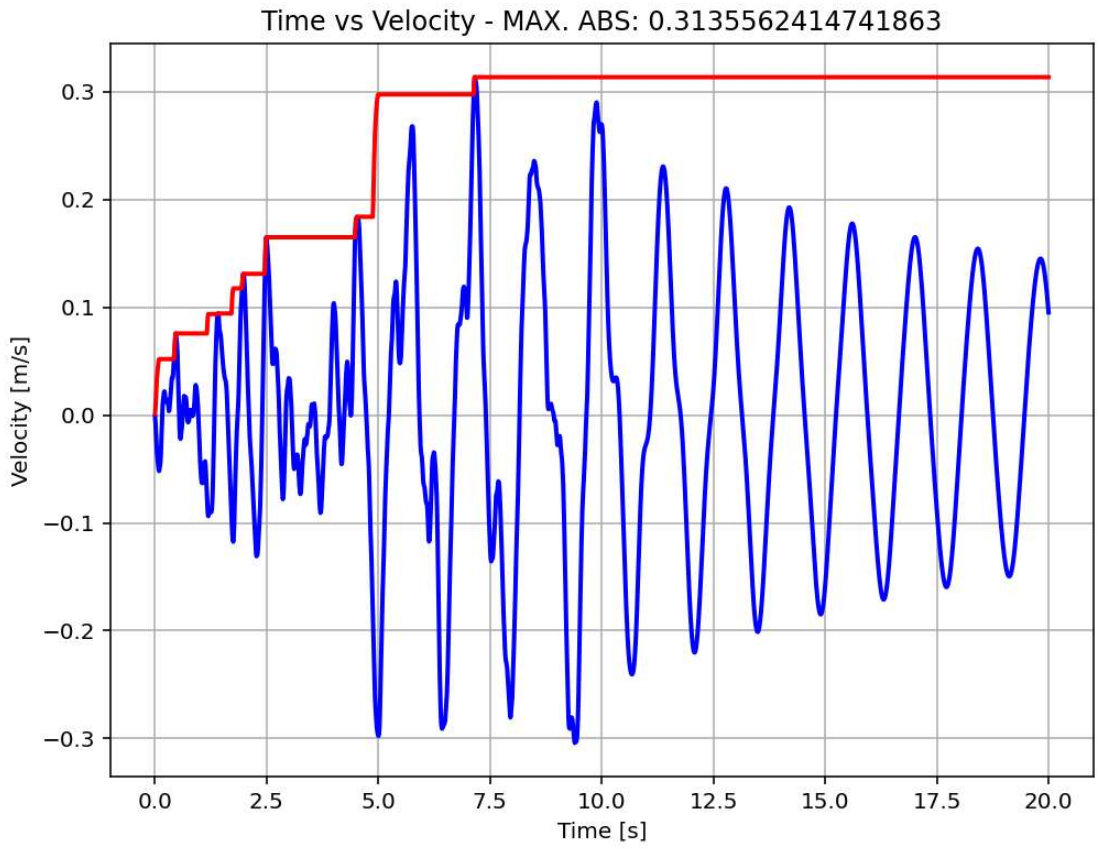


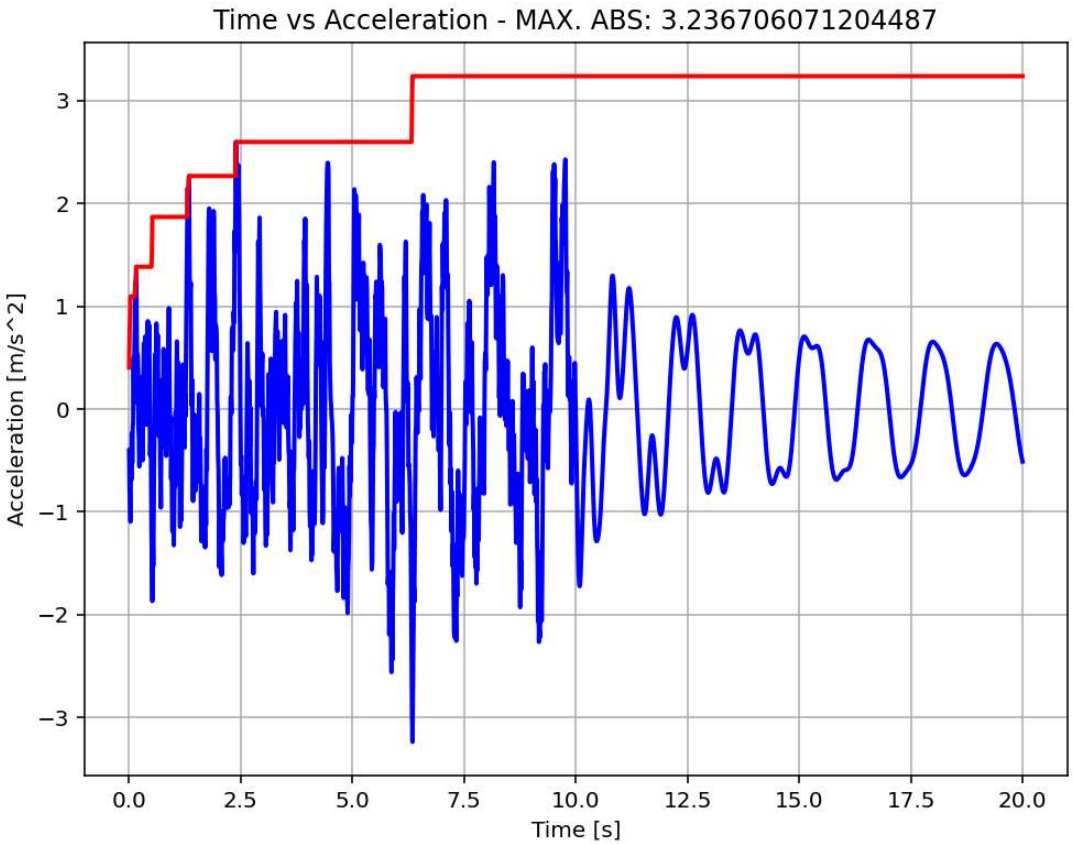
$$Energy\ Dissipation\ Capacity\ Index = \frac{E_d(Dissipated\ Energy\ from\ Earthquake)}{E_d(Dissipated\ Energy\ from\ Cyclic\ Displacement)} * 1$$

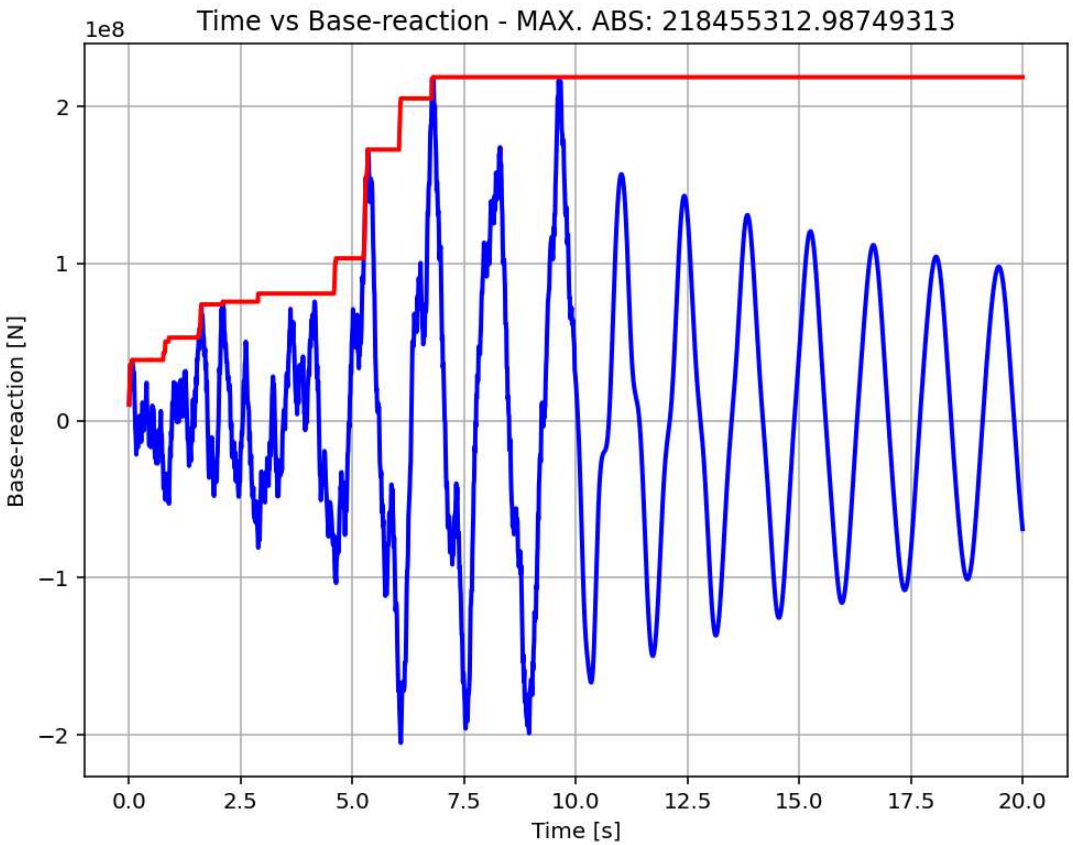

Fragility Curves
$\xi_{eq} = \frac{E_d}{4\pi E_s} * 100$
Minor Damage Level (η=20.0, β=5.0)
Moderate Damage Level (η=40.0, β=5.0)
Severe Damage Level (η=60.0, β=5.0)
Failure Level (η=100.0, β=5.0)
Probability of Exceedance
10^0
6×10^{-1}
4×10^{-1}
3×10^{-1}
0
20
40
60
80
100
Equivalent viscous damping ratio (%) [IM]

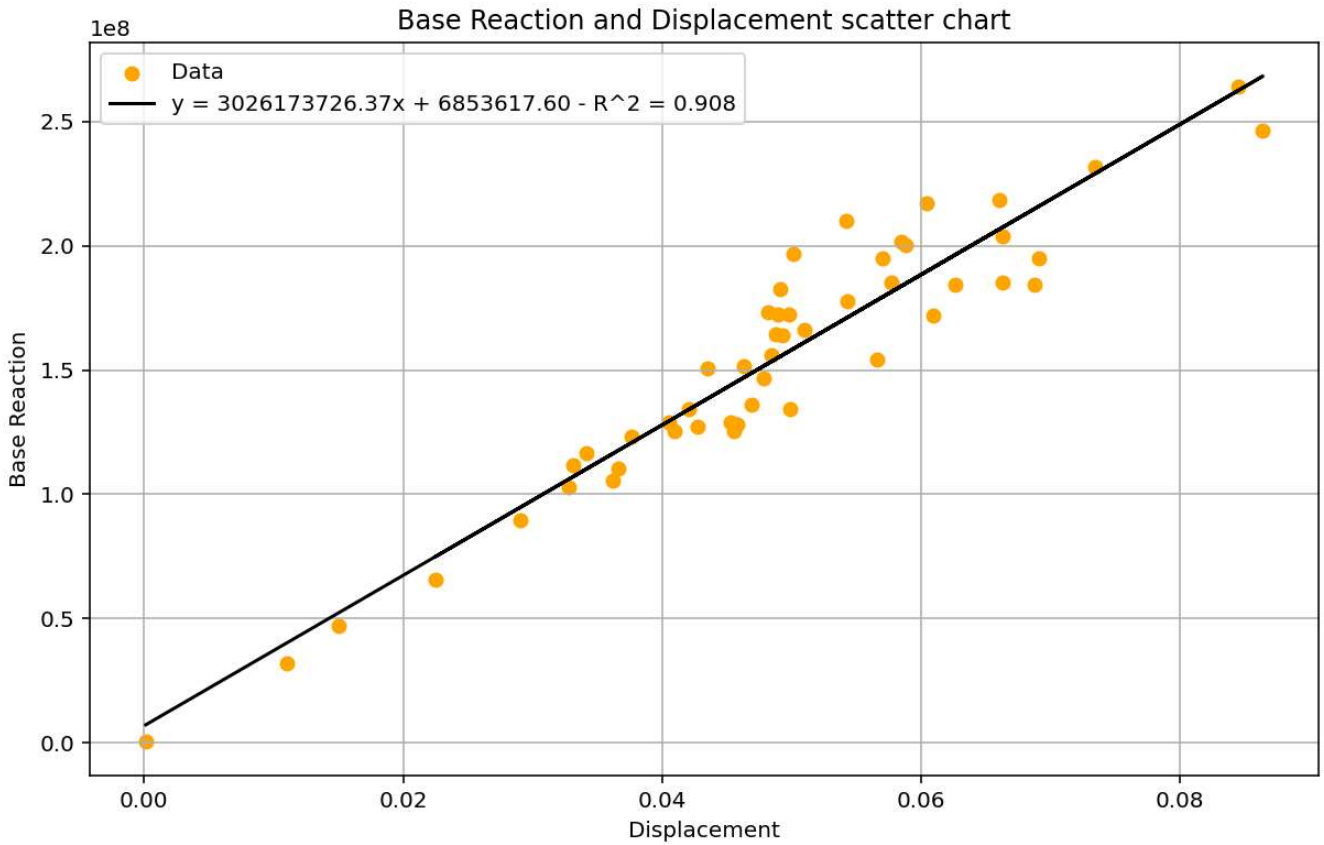

Displacement & Base Reaction Relation From Last Dynamic Analysis
1e8
2
1
0
−1
−2
Base Reaction [N]
−0.06
−0.04
−0.02
0.00
0.02
0.04
Displacement [m]

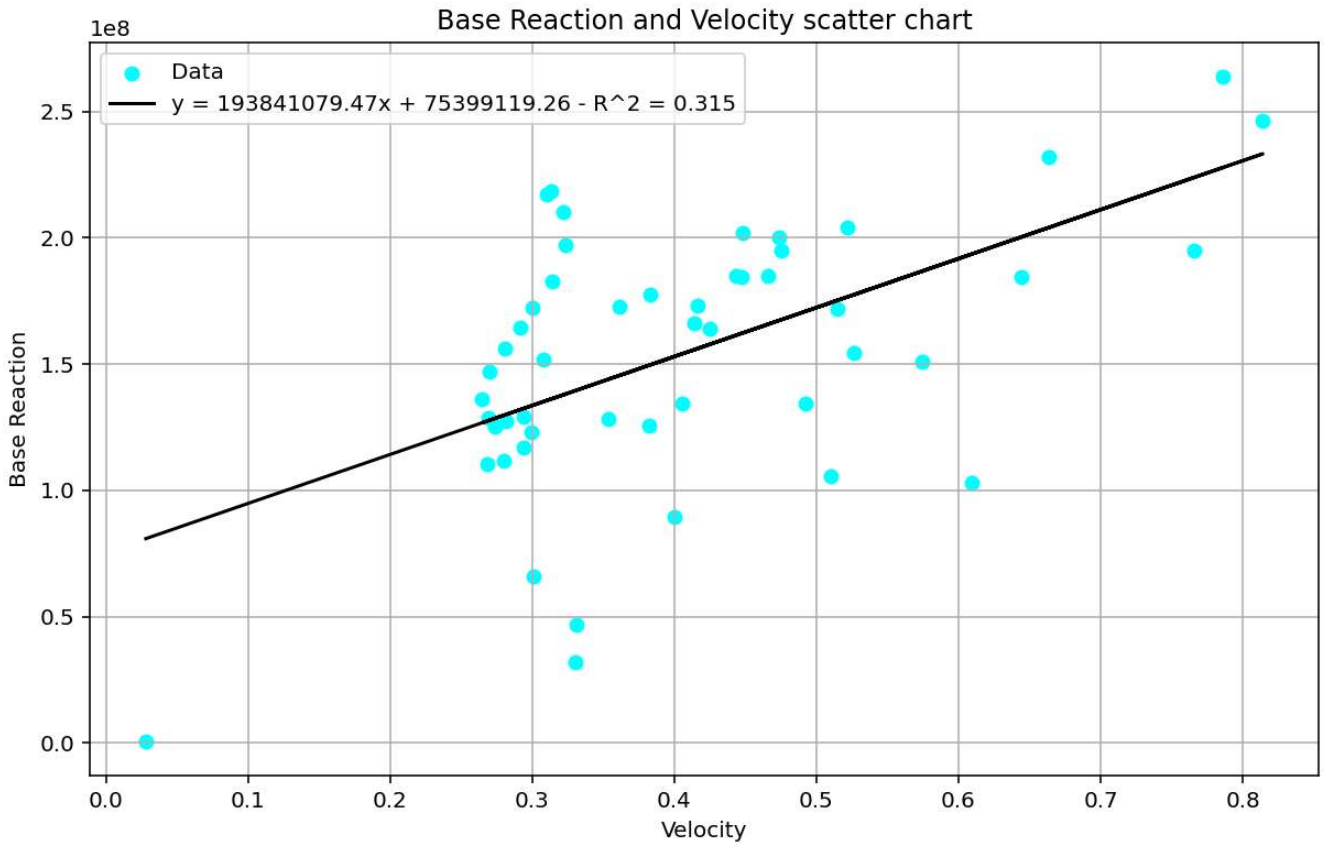

Time vs Displacement - MAX. ABS: 0.06604102293072457
Displacement [m]
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

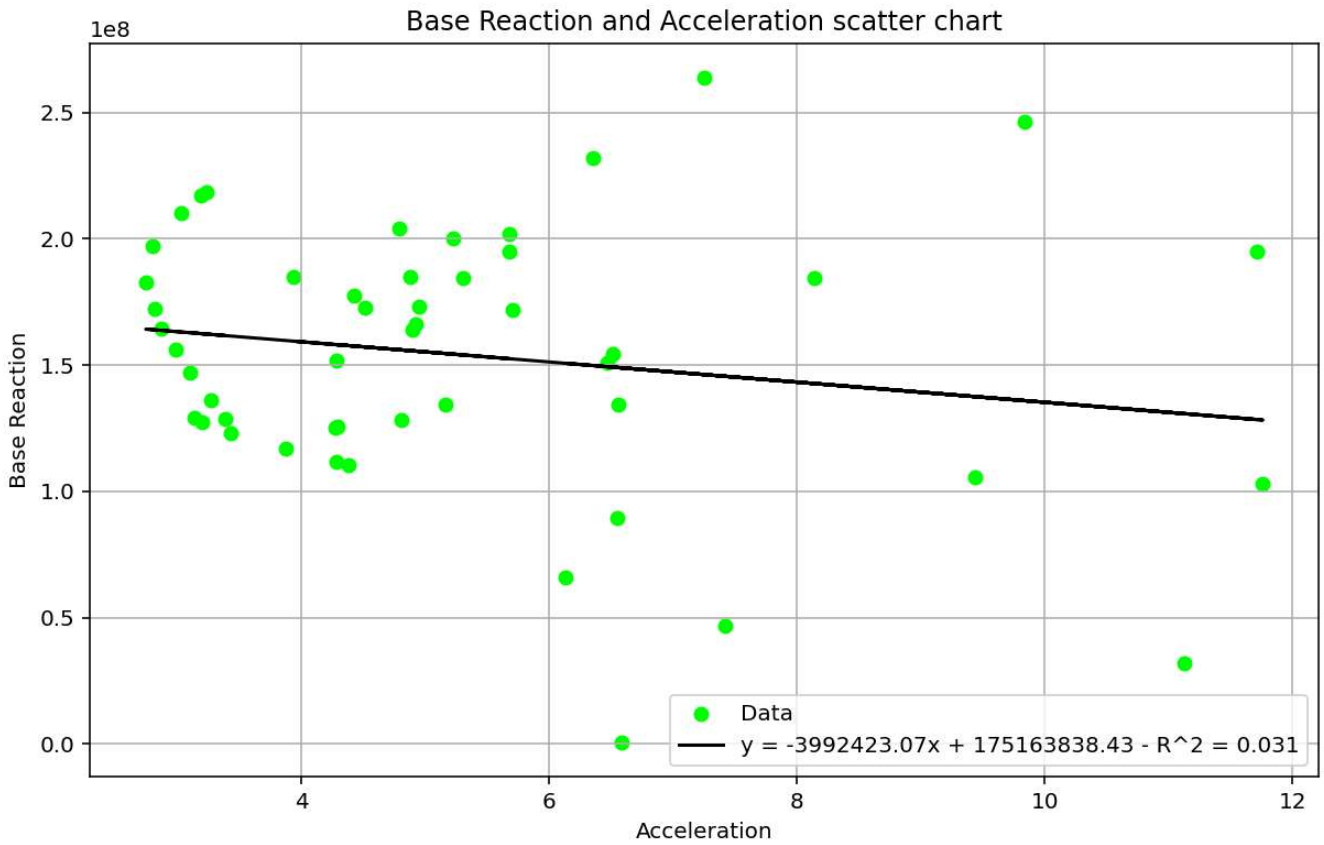

Time vs Velocity - MAX. ABS: 0.3135562414741863
Velocity [m/s]
Time [s]
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

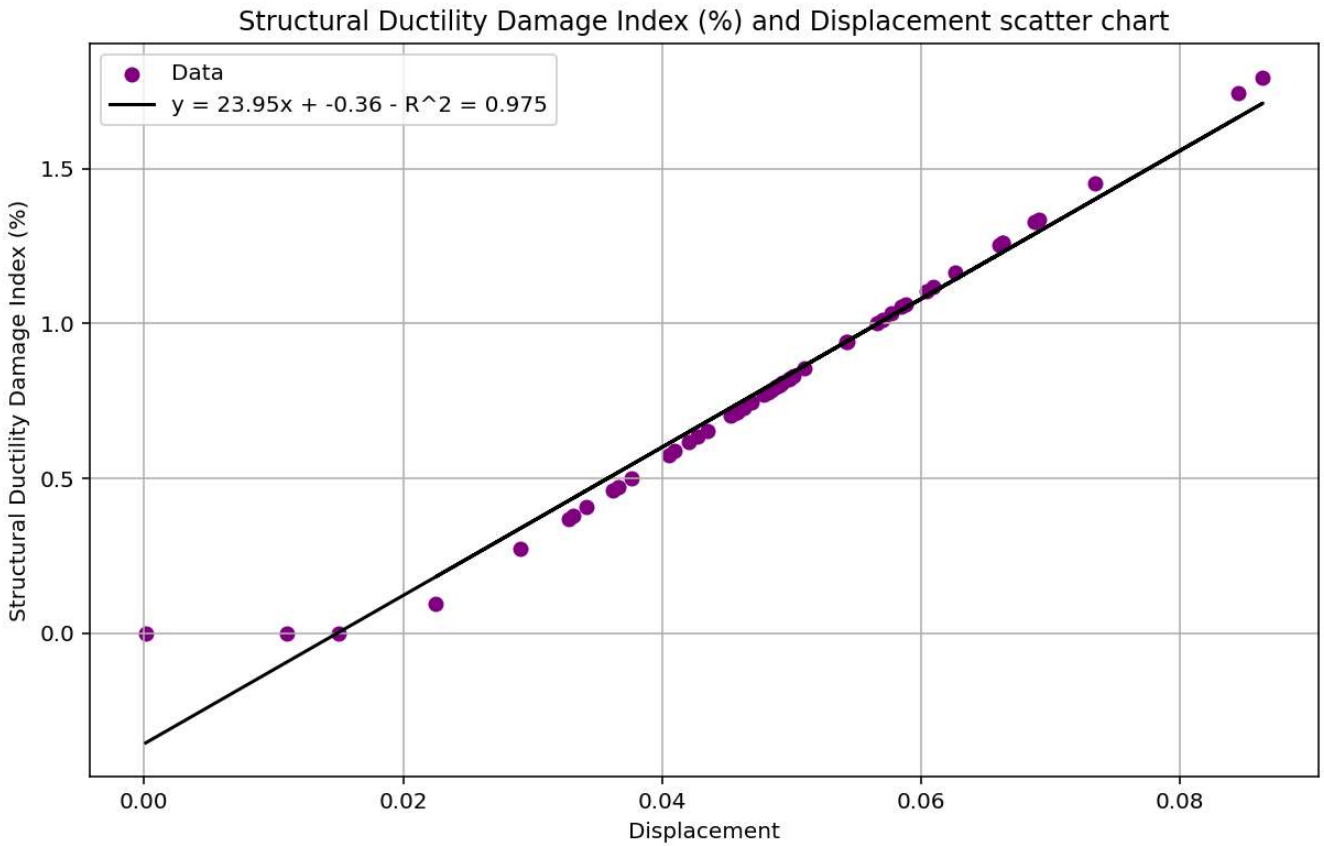

Time vs Acceleration - MAX. ABS: 3.236706071204487
Acceleration [m/s^2]
Time [s]
3
2
1
0
−1
−2
−3
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

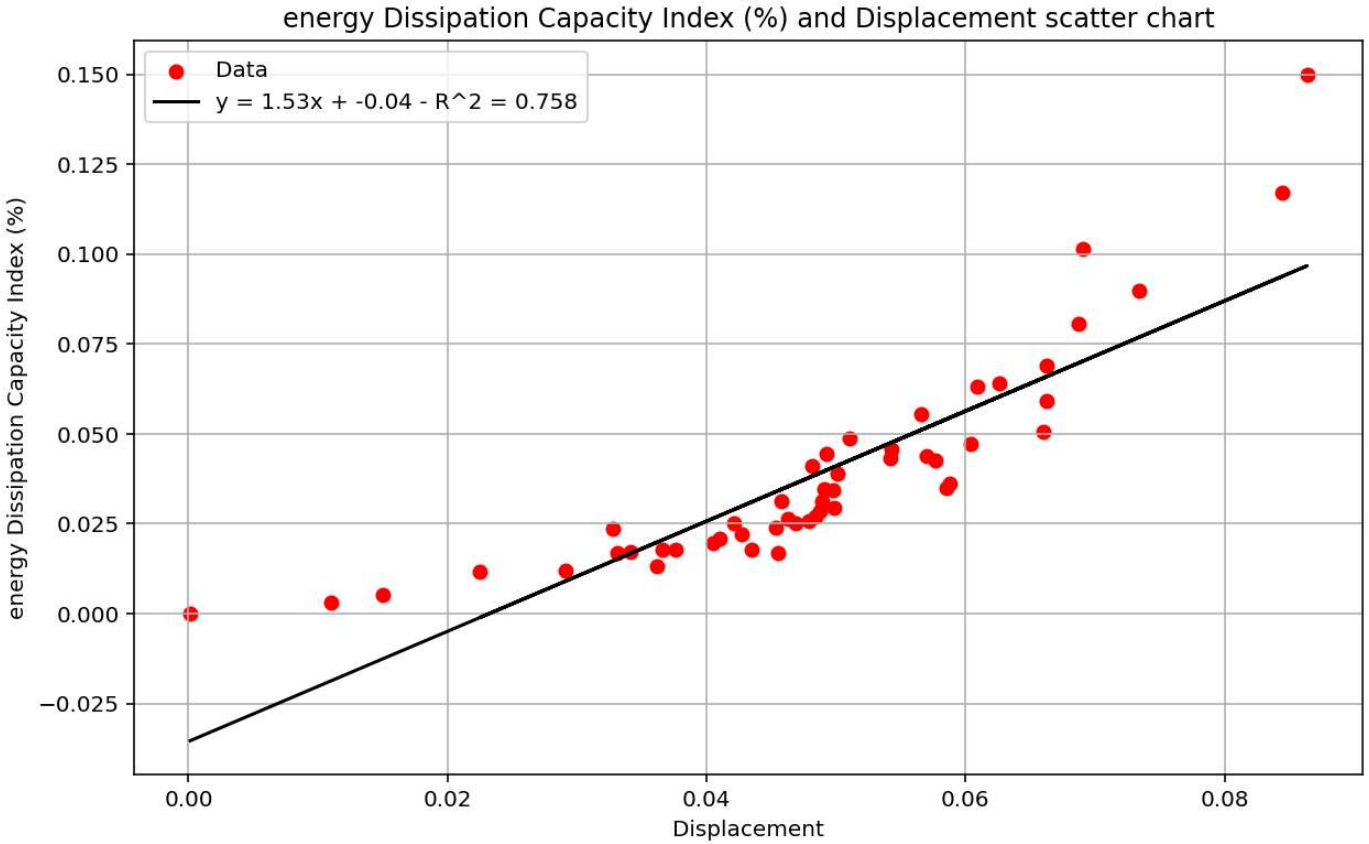

Time vs Base-reaction - MAX. ABS: 218455312.98749313
1e8
2
1
0
−1
−2
Base-reaction [N]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


Base Reaction and Displacement scatter chart
1e8
Data
y = 3026173726.37x + 6853617.60 - R^2 = 0.908
Base Reaction
Displacement
0.0
0.5
1.0
1.5
2.0
2.5
0.00
0.02
0.04
0.06
0.08

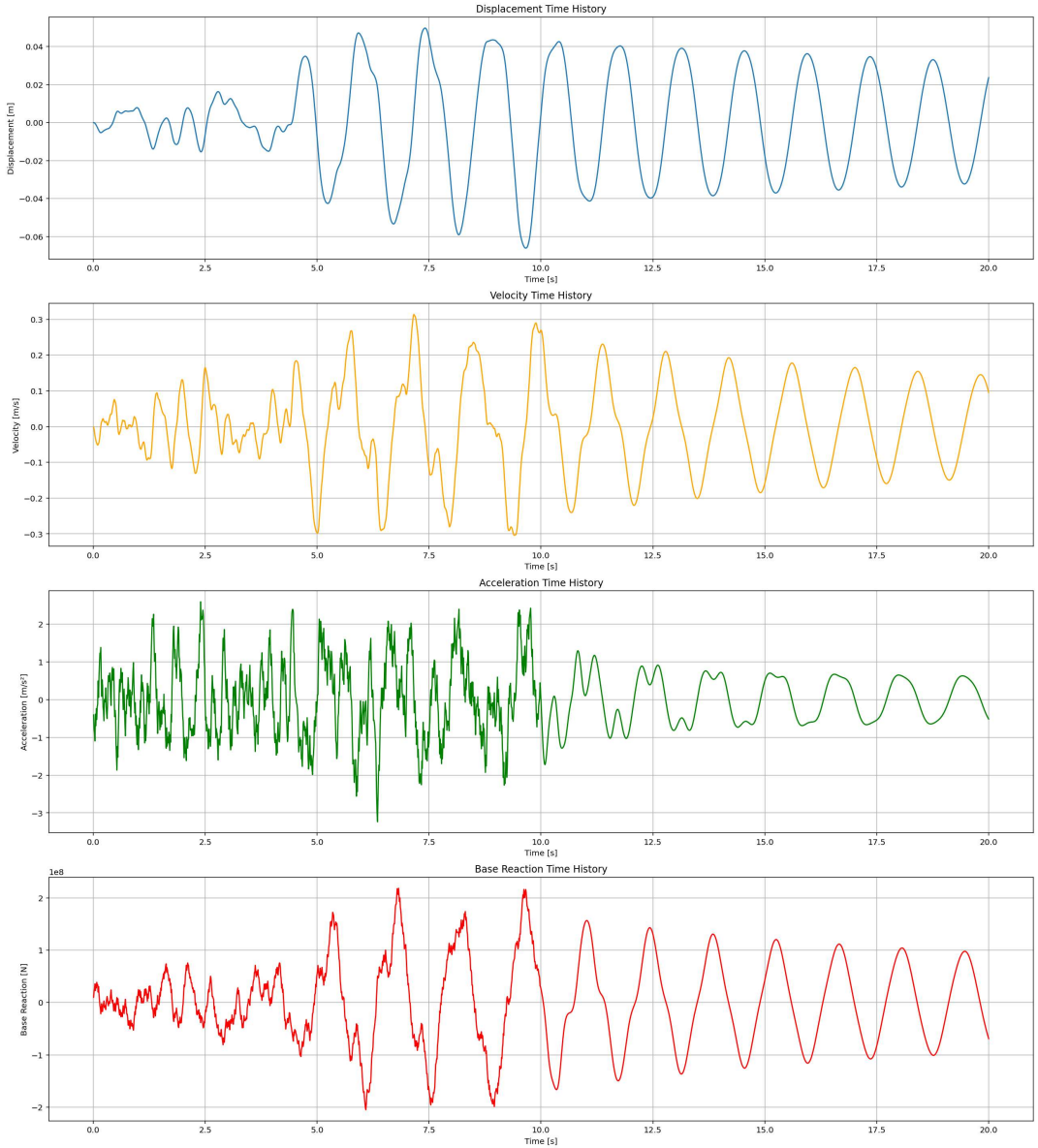

Base Reaction and Velocity scatter chart
1e8
Data
y = 193841079.47x + 75399119.26 - R^2 = 0.315
Base Reaction
Velocity
0.0
0.5
1.0
1.5
2.0
2.5
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8

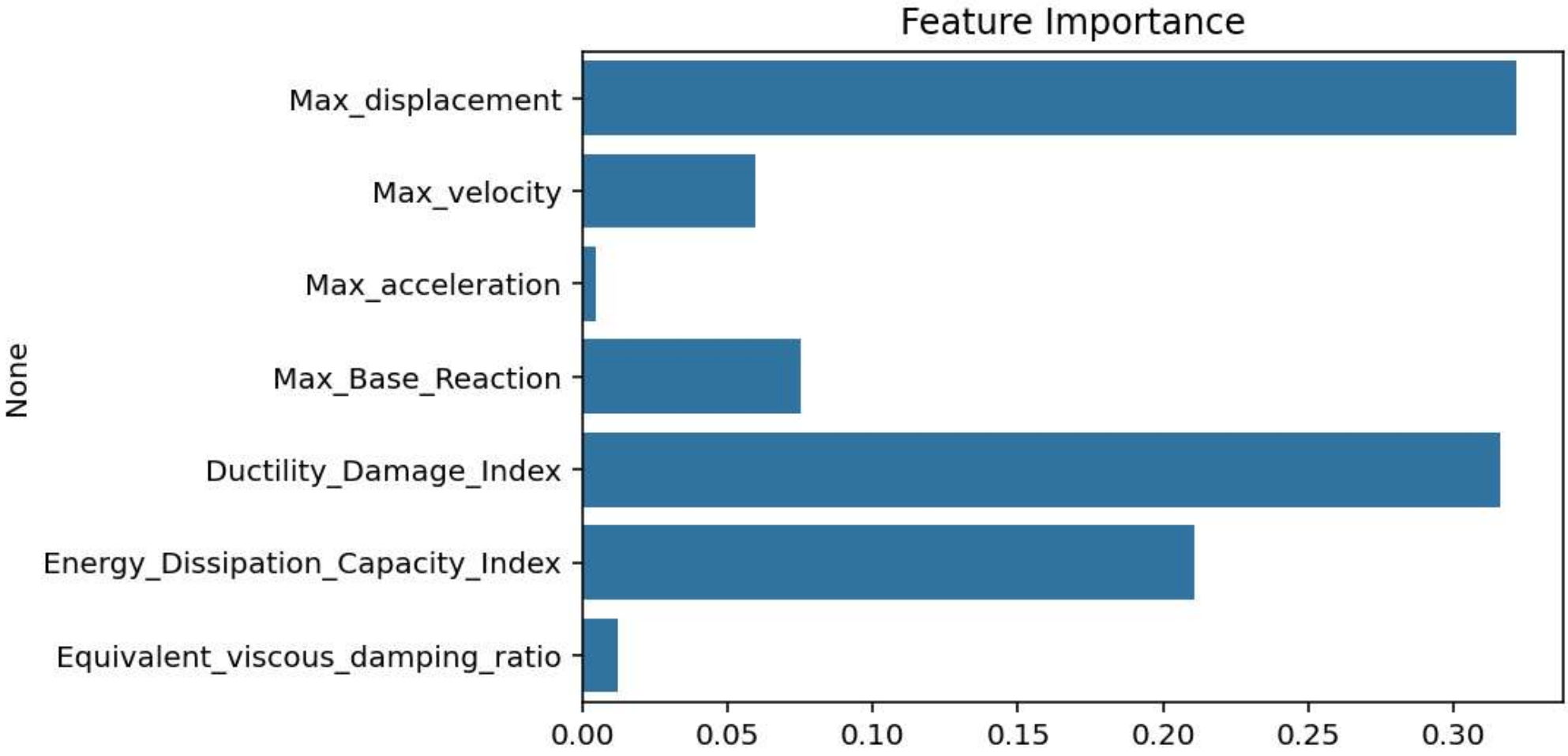

Base Reaction and Acceleration scatter chart
1e8
2.5
2.0
1.5
1.0
0.5
0.0
Base Reaction
4
6
8
10
12
Acceleration
Data
y = -3992423.07x + 175163838.43 - R^2 = 0.031

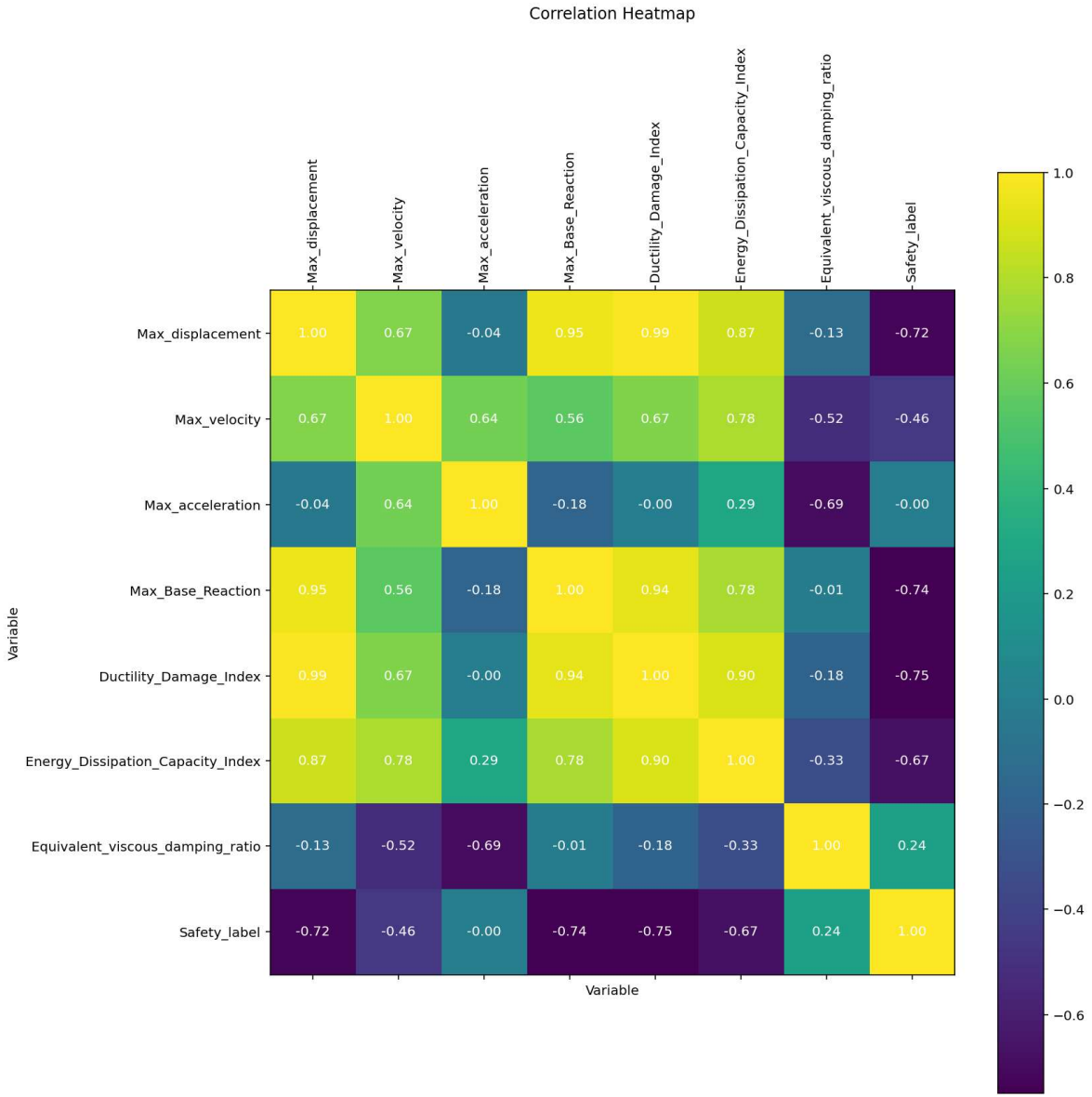

Structural Ductility Damage Index (%) and Displacement scatter chart
Data
y = 23.95x + -0.36 - R^2 = 0.975
Structural Ductility Damage Index (%)
1.5
1.0
0.5
0.0
0.00
0.02
0.04
0.06
0.08
Displacement



$$Energy\ Dissipation\ Capacity\ Index = \frac{E_d(Dissipated\ Energy\ from\ Earthquake)}{E_d(Dissipated\ Energy\ from\ Cyclic\ Displacement)} * 100$$


Displacement Time History
Displacement [m]
Time [s]
Velocity Time History
Velocity [m/s]
Time [s]
Acceleration Time History
Acceleration [m/s²]
Time [s]
Base Reaction Time History
1e8
Base Reaction [N]
Time [s]


Feature Importance
Max_displacement
Max_velocity
Max_acceleration
Max_Base_Reaction
Ductility_Damage_Index
Energy_Dissipation_Capacity_Index
Equivalent_viscous_damping_ratio
None
0.00
0.05
0.10
0.15
0.20
0.25
0.30


Correlation Heatmap
Max_displacement
Max_velocity
Max_acceleration
Max_Base_Reaction
Ductility_Damage_Index
Energy_Dissipation_Capacity_Index
Equivalent_viscous_damping_ratio
Safety_label
Variable
Variable
1.00 0.67 -0.04 0.95 0.99 0.87 -0.13 -0.72
0.67 1.00 0.64 0.56 0.67 0.78 -0.52 -0.46
-0.04 0.64 1.00 -0.18 -0.00 0.29 -0.69 -0.00
0.95 0.56 -0.18 1.00 0.94 0.78 -0.01 -0.74
0.99 0.67 -0.00 0.94 1.00 0.90 -0.18 -0.75
0.87 0.78 0.29 0.78 0.90 1.00 -0.33 -0.67
-0.13 -0.52 -0.69 -0.01 -0.18 -0.33 1.00 0.24
-0.72 -0.46 -0.00 -0.74 -0.75 -0.67 0.24 1.00
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6